Contents

A Field–Matter Selector for Outcome Production, Conditioned on a Detector Ready-State Measure

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[DRAFT v0.9 — base v0.5 of 2026-07-28 with point revisions through 2026-07-29, itemized below (header label harmonized 2026-08-01, tri-paper consistency ledger item 30). Post-review revision, deliberately separate from v0.4 so the review record stays clean: v0.4 is the consolidated response to the 2026-07-27 AI review panel (GPT-5 Codex / Gemini 1.5 Pro / SuperGrok; internal Fable 5 down-weighted) and is frozen with that record; the panel did not see the material added here. v0.5 adds two items that emerged in post-review discussion between the authors: (a) a physical-origin sketch for the P2 noise scaling — the interference cross term between each site's complex amplitude and the uncontrolled-phase fields of the detector's own charges and the vacuum (§4, with the homodyne limit as consistency check and a corresponding entry in §9.4); (b) the explicit statement that capture is reversible, with echo and quantum-memory experiments as evidence (§1, §2, §3). The v0.5.1 touch-up (same day) develops (a) one step further: the unknowability of the surface fields restated as a universality argument; the companion terms of the same expansion (bath heating, additive noise floor, damping) tracked, yielding the fairness window $\hbar\omega \gg k_BT$ with the detector-family consistency check (§4); and a warm-detector odds-bias candidate* flagged in §8.6(v), pending systematics analysis. The v0.5.2 addition (same day): §8.7, the cross-station injection test for working Bell setups, built on the two-station correlated-noise calculation (marginal protection; the white-noise protection theorem; the registry-infidelity leak law $\delta S = 4(1-\eta)L$; the phase-locked-tone channel with its above-Tsirelson signature, and the single-station fixed-phase calibration of $\kappa\lambda$ that makes a two-station null falsifying rather than merely disappointing), with §7.2 and §9.4 hooks and the simulation suite referenced in Appendix D. The v0.5.3 revision (2026-07-28, same day) is a cross-round correction plus two labeled interpretations: finding N2 of the *companion measurement-structure paper's* review panel exposed a misreadable identification in Theorem 2 — the deficit rate $\Delta E/\hbar$ is renamed κ_{ret} (a reversible return rate to the common field, not bath decay; dictionary paragraph added in §5.2), its ansatz status is stated plainly, and the microscopic derivation is logged as open problem §9.4(viii); and two sponsor-originated interpretive passages are added, explicitly labeled as organizing readings rather than derivations — the diffused-hologram account of the noise (§3 control case; §4 paragraph: reference incoherence as the common origin of fairness and irreversibility; the phase history converted into the tie-breaker noise; the slaved phase as memory). The v0.5.4 addition (2026-07-29): §8.6(vi), *energy-audit interferometry* — a proposed consistency test (deposited-vs-created energy per click on an energy-resolving detector plane behind an interferometer; full quantum per click, no fractional deposits), with a sponsor-originated interpretive clause (the distributed Stage-1/2 excitation is reversible, so

losing sites retain no calorimetric residue; deposition completes only at registration) and its §9.4(viii) hook; a literature search found no experiment framed as this audit — the canonical interference experiments used energy-blind click detectors, and the closest data is incidental (energy-resolving MKID astronomical cameras; TES calorimetry). The v0.5.5 revision (2026-08-25) promoted §4's first knife-edge failure mode to a named alternative, *the rogue wave*, stated at its strongest before being refuted; Grangier et al. 1986 was added to the references, supplying the citation for the beam-splitter anticorrelation invoked in §3 and §8.6(vi).

The v0.6 revision (2026-08-31) is a de-claiming pass, not a new result. It applies the editorial changes required by the quantum-equilibrium selection revision plan and its two independent reviews, which found the previous draft "directly contradictory and therefore non-authoritative" against that plan's claim boundary. Nothing in §§3–8 changes technically; what changes is what the paper says it has shown. Specifically: (i) the title no longer claims a *derived fair game*; (ii) the abstract now opens by separating the two probability spaces — the quantum outcome measure adopted as comparator, and the detector ready-state preparation measure held as an explicit premise — and states that this paper supplies at most the selector half of a larger theory; (iii) the claim that "the Born weight is nowhere inserted by hand" is **withdrawn** in §1, replaced by the narrower true statement, with P0+P1 identified as the single place outcome weights enter; (iv) "the measurement postulate is redundant" is replaced throughout by conditional outcome compatibility over a tested domain (§1, §2, §9); (v) the premises' "standard detector physics" lineage is distinguished from their epistemic status — the deposited-energy law and P4(ii) commitment-rate structure are marked as *prior model components*, not established event-level microscopic facts (§2); (vi) Robert Adler's injection-locking equation is disambiguated from Stephen Adler's trace-dynamics and collapse work at every use, and R. Adler 1946 is added to the references; (vii) the rogue-wave framing is removed from §4 — the underlying extreme-value/threshold failure result, its exceedance scaling, and the causal-inversion diagnosis are retained in compressed form, since §5 depends on them; and (viii) §1 and §9 now state that exclusivity, quench, and energy routing — the production of exactly *one* record — are owed by the selection dynamics and are not discharged anywhere in this paper.

One technical correction (2026-08-31), in P3(b). The residual off-diagonal noise correlation was stated as $C = \mathbb{1} + O(f)$ with $f = \Gamma_{\mathrm{rad}}/\Gamma_{\mathrm{tot}} \sim 10^{-6}$. That is the f-dominated branch and is not binding at the quoted value of f; under the fluctuation–dissipation theorem the correct scaling is $C = \mathbb{1} + O(\max(f, e^{-d/\ell}))$, with d the site spacing and ℓ the local scattering correlation length. This surfaces a geometric condition the premise had not stated, $d \gg \ell$, which is unverified for the detector families discussed in §6.1 and §8 and is now flagged as open. Verified numerically in `code/check_g1g2_ready_state.py`.

The v0.7 revision (2026-08-31) fixes an internal inconsistency about continuum absorbers. P4(ii) said registration in class (ii) is "a rate process operating throughout selection" — a fast-commit description — while §6.1's timescale ladder puts commitment three to five orders *slower* than the game. Both cannot hold. §6.1's ladder is adopted: both registration classes reach an outcome by first passage (Theorem 0), and Theorems 4–5 are demoted from the operative mechanism for class (ii) to a robustness result covering the counterfactual fast-commit regime. This is a strengthening — the paper no longer rests on Theorem 5's linearity claim, which the continuum-detector literature does not support (exponential SNSPD efficiency at low photon energy; measured supralinear response in SPADs and SNSPDs). Verified by simulation separating commit speed from commit law (`g3_drain_tests/theorem5_check.py`): a deliberately wrong Arrhenius commit law gives 0.719 against Born's 0.500 at fast commit and 0.504 at §6.1's separation. Additionally, P4 gains an explicit clause (a): **registration consumes the full quantum**. That premise was previously implicit in Theorem 5's accounting, and the exclusivity argument depends on it — with a closed pot, a full-quantum threshold makes two simultaneous registrations energetically impossible, so exclusivity follows from conservation rather than from an imposed stopping rule.

Also in v0.7: the P5/P6 quarantine claim is withdrawn (§2, third remark). Earlier drafts asserted that no single-detector result uses P5 or P6. The counterexample is internal: §3 assigns Stage 2 the job of supplying beam-splitter anticorrelation, and under a closed pot with the full-quantum threshold of P4(a), that pot must be closed across two spacelike-separated arms — otherwise a balanced beam-splitter leaves $\hbar\omega/2$ in each arm, neither reaches threshold, and neither ever registers. The single-detector sector therefore consumes a nonlocality premise whenever the incident quantum's support spans spacelike-separated registration sites. P5 as written did not cover that case (it was stated for multi-quantum and multi-partite states), and its scope is broadened in this same revision, in one pass with §7's fidelity law. That broadening adds no free parameter: in the single-quantum sector a surviving far-arm fraction would create energy against the closed pot, so $\eta = 1$ is forced there and exact anticorrelation is predicted outright — which also supplies a second, conservation-based reason for §7.3's geometric argument that $\eta = 1$ exactly. This is a genuine increase in the cost of the single-detector theory, recorded rather than minimized — though it is not peculiar to this mechanism, since any single-world account of anticorrelation carries a nonlocal constraint.

The v0.8 revision (2026-09-02) corrects the timescale ladder of §6.1 against the paper's own definition of commitment, and withdraws in turn the v0.7 demotion of Theorems 4–5. §6.1 quoted $\tau_{\text{commit}} \sim$ ns–μs for a silicon SPAD. That figure is the avalanche-and-quench cycle — record-writing — whereas commitment in the sense every other section uses (Theorem 2, §4's diffused hologram, §3, §8.6(vi)) is *first irreversibility*: the moment the deposited excitation ceases to be coherently returnable to the common field. For a gapped absorber that clock is protected only while a site's share lies below $\theta = E_{\text{gap}}/E_{\text{photon}}$; above θ a real electron–hole pair is available and the interband vertex plus thermalisation, 10 fs–1 ps in silicon, is the first irreversibility. Recomputed on that basis (`code/silicon_commit_recompute.py`; reconciliation ledger E-16), the top rung is *marginal* for silicon — the leader's exposed climb from θ to 1 takes 13–84 fs against a 10 fs–1 ps commit, a ratio straddling unity — and *inverted* for superconducting absorbers ($\theta \approx 0.002$; an exposed leg of 0.8–1.6 ns against a 10–50 ps hotspot). The middle separation $\tau_{\text{game}}/\tau_c = 2\ln^2 N$ is Γ-invariant, a robustness result now stated. Consequences: (i) §6.1's "two to four orders of magnitude to spare at every rung" is withdrawn; (ii) the v0.7 claim that first passage settles every outcome before any commit-rate law acts is withdrawn, and Theorems 4–5 return to load-bearing status for continuum absorbers, with the linearity Theorem 5 must supply identified as that of the first-irreversibility vertex, not of the downstream cascade; (iii) P4(a) is marked as a premise about the quantum wherever $E_{\text{photon}} > 2E_{\text{gap}}$ (silicon below 553 nm; all superconducting absorbers), since there the closed pot could pay for two real sub-gap excitations and the gap does not enforce exclusivity; (iv) a threshold-gated commitment channel is added to Table 1 as a live candidate, with a gap-ratio-graded test in §8.6(vii) and the share-ontology question in §9.4(xi). This revision costs the paper margin it had claimed. It is recorded because the definition was fixed before it was applied, and applied where it hurt.

The v0.9 revision (2026-09-02) adopts reading B of the share ontology as the working assumption, records it as a choice, and restates the ladder and the role of Theorems 4–5 accordingly. v0.8 left a fork open in §9.4(xi): are the shares energies that would decay if they could (reading A), or amplitude bookkeeping that becomes energy only when a site takes the whole quantum (reading B)? Two results decide the working choice. First, the only microscopic model in this program (`first_mark_two_absorber/`, a 576-state field–absorber–bath Hamiltonian) enforces B by construction — the absorber's reversible excitation carries the full quantum, and what is split between arms is amplitude — and standard QED agrees at the vertex. Second, reading A, quantified in the paper's own discrete engine (`g3_drain_tests/threshold_gated_commit.py`), does not survive it: the eventual winner spends 40–90% of the game above θ, not the one exchange step the v0.8 continuum estimate gave, so the threshold-gated channel is fully open in silicon rather than marginal; and a gated commit law — linear or not — biases an 80/20 split by +0.04 at one expected commit per exposed leg and by +0.12 to +0.19 at ten, because Theorem 4 requires the conditional pick to equal the share for *every* site, gated or not. Heralded 505 nm data taken with silicon SPADs ($g^{(2)}(0) = 0.0023$) already bound the no-P4(a) version of A four orders below the marginal range. Under B: commitment is available from every site at the golden-rule rate, linear in the site's holding (Theorem 5), and Theorem 4 gives the Born weights at any commit speed. Consequences: the top rung of §6.1's ladder is no longer a condition; the exchange game is the reversible sub-threshold dynamics of Theorems 1–3 and carries no statistical weight of its own; and the claim moves from "the game realizes the Born measure" to "any actualization law whose hazard is linear in the share reproduces the Born comparator, and the fairness theorems state what the substrate must not do to the shares before it fires." The threshold-gated channel (Table 1), the gap-ratio grading (§8.6(vii)) and the SPAD/SNSPD asymmetry are retained as what reading A would predict, with their bounds, so the fork stays testable. What the substrate must now supply is the actualization law itself — which site fires, in one world, with hazard proportional to its share, without that proportionality being inserted — and that is the question the framework companion's Adler-locking race is built to ask. *First answer, at diagnostic budget (2026-09-02, `adler_two_channel_exploratory/`; the package's frozen numerical budget is unmet, so this is not a* *production result):* a nine-angle polarization sweep at the frozen criterion gives an unconstrained coupling exponent of 1.56 (95% $[1.44, 1.69]$), rejecting both the amplitude-linear and the amplitude-squared curves; the exponent is criterion-dependent, rising from 1.4 to 1.8 ($[1.60, 1.96]$, Born still rejected at nine cells) as the fixed dwell doubles, and rising likewise as the pulse shortens or the noise strengthens — i.e. it moves toward Born only as a substantial fraction of trials produce no commitment at all, and at twice that dwell again the efficiency collapses to 18% before the exponent is determinable. As it stands the race does not supply a universal hazard linear in the share. The plan's labelled positive control, a dwell inversely proportional to the coupling, was then run at the sponsor's request and does not rescue it: the exponent overshoots to 3.81 $[3.50, 4.15]$, so the Born exponent lies between the amplitude-neutral criterion and the scale-similar one and is reached by neither — the plan's own "no finely tuned criterion" falsification row, met from both sides — and a tuned dwell $\propto K^{-1/4}$, run as a demonstration, reproduces the curve exactly ($p = 2.00$ $[1.84, 2.16]$), which quantifies the tuning (the exponent must be held to 0.25 ± 0.07) and is a fitted criterion, not a mechanism. The lock-tolerance and spectral sensitivities were then run at the same budget: the exponent stays between 1.6 and 1.8 across tolerances of 0.175–0.70 rad, rising only where commitment becomes unreliable; and across Gaussian, Lorentzian, peaked and notched detuning densities the direct exponent moves with the spectrum in the same order as the analytic comparators, sitting about one half power above the eligible-count exponent rather than the full power the rate-weighted flux would give — the race scales as tongue width times the square root of the rate. The source of that half power was then computed: the race's deterministic skeleton, with no noise at all, reproduces the exponent (1.52 against 1.56), because each clock's entry is a near-deterministic Adler slide from its random starting phase and the fastest of N such slides gains only logarithmically in N, where a Poisson race — the rate-weighted flux, the semicircle, the square — would gain a full power. A Born-producing race would need each site's commitment to be memoryless, an exponential waiting time with hazard proportional to the local relaxation rate, so that the hazards add across the tongue; that requirement is the golden rule stated in the substrate's language, and it is what reading B imports. That rule was then run as a variant at the same budget: with commitment made memoryless and its hazard linear in the energy a clock absorbs inside its tongue, using the Adler clock's own power law $K \cos \theta$ and no inserted square, the same clocks, grid and pulse give an exponent of 2.16 $[2.14, 2.19]$, within about 0.02 of the Born curve at every angle, tracking the time-integrated tongue flux ($\propto K^{2.2}$ on that grid) — the plan's rate-weighted mechanism realised, with the square supplied by tongue width times locked absorption rate rather than inserted. What that variant puts in by hand is the golden rule's structure, memorylessness and hazard linearity in energy, i.e. Theorem 5 and reading B; what the substrate supplies is the amplitude dependence. Whether a detector's commitment process has that structure is the E-16 question — first irreversibility as dissipation into a continuum. A lagged-hazard sweep then showed the result is indifferent to the hazard's *timing* (exponents 2.16–2.21 for hazard memory from zero to ten commitment times, since a lag common to every site reparametrizes time for both channels alike) and sensitive only to its *form*, so what a detector must supply is a stochastic commitment linear in absorbed energy with a site-independent time profile — the absorption vertex's own structure in both detector families, with the deterministic cascade downstream of it (avalanche, hotspot, quench) carrying no which-site weight. On that reading the SPAD/SNSPD asymmetry does not arise from this mechanism, and the picture is reading B's. Whether the vertex's stochastic selection is one-world actualization or a rate over an ensemble is then decided by exclusivity: independent site rates predict both ports of a balanced split committing in essentially every trial, a coincidence ratio of 1 against the heralded-photon value of 2×10^{-3}, and a one-world stop reproduces the data only if it acts within about 10^{-3} of the commitment time — for silicon, within a femtosecond or less, in which light travels less than a micron, against millimetre-to-metre port separations. The selection is one-world, and its exclusivity is the nonlocal closed-pot constraint P5 that §2 already concedes the single-detector sector consumes; nothing in these simulations derives it. The production budget remains unrun.

Still pending: final byline; the full structural revision to the plan's eleven-section manuscript architecture, which remains gate-blocked.]*

Contributions. Claude Fable 5 (Anthropic) performed the formalization, the theorem statements and proofs, the simulations and their correction, the literature work, and the manuscript prose. John M. Bramble, MD supplied the physical framing and research direction, adjudicated scope and interpretation, and is the accountable human sponsor. Byline order is the sponsor's decision and may be revised before any submission; `CITATION.cff` is to be reconciled to match.

Abstract. *This paper supplies, at most, the selector half of a larger theory. It does not derive the Born measure, and the reader should hold it to the narrower claim stated here.* Two probability spaces must be kept apart throughout: the **quantum outcome measure** $B_\Psi^M(a) = \langle\Psi|\Pi_M(a)|\Psi\rangle$, which we adopt as a comparator from the structural literature and do not derive; and a **microstate preparation measure** over the detector's own ready state, which is an explicit premise of this paper with independently measured provenance, not a result of it. The Born rule is the sole entry point of probability into quantum mechanics, and it is postulated: structural derivations (Gleason, envariance, decision theory, typicality) establish uniqueness of the measure without supplying a selection process, while dynamical-reduction models supply a process only by engineering their noise so that squared amplitudes evolve as a fair game. We present a candidate mechanism in which the game is physical and the fairness follows from stated premises about detectors rather than being engineered. The result is deliberately conditional and it does not derive the square: that the measure can only be Born is Gleason's theorem, and that a driven oscillator's deposited energy is quadratic in amplitude is classical energetics — neither is ours to claim. What we offer is a candidate *selection process* between them, and what we test is whether pushing the frozen preparation measure through that process yields an outcome law compatible with the comparator — **conditional outcome compatibility**, which is neither a derivation of the Born measure nor a statement that any equilibrium is dynamically preserved. An incident quantum deposits energy across all candidate absorber sites in proportion to squared local amplitude — driven-oscillator energetics, the only place the square enters — and the deposited energies then compete through noise-driven exchange until one site registers the quantum; a classical martingale theorem (Pearle's gambler's ruin) then equates registration probabilities with initial energy shares, i.e. Born weights, given the premises. Four theorems replace the engineered-noise postulate: amplitude-linear coupling forces the unique drift-free noise scaling; sub-threshold absorbers possess no autonomous phase, eliminating winner-take-all feedback, with level discreteness sharpening the threshold; the locality of detector noise protects site independence, with the penalty for correlation computed exactly; and energy conservation makes registration rates kinematically linear in occupation, which — in the counterfactual regime where commitment is fast enough to compete with the game — renders the statistics exact at every registration speed and in every quantum sector. In the timescale regime real detectors actually occupy (§6.1), commitment is slow and the outcome is fixed by first passage before any commit-rate law can act; the two registration classes deliver Born by one corollary, not two. A mechanism sketch grounds the required noise scaling physically: the energy fluctuation of a site is the interference cross term between its complex amplitude and the uncontrolled-phase wave packets of the detector's other charges and the vacuum field — zero-mean because the detector holds no phase reference, amplitude-linear because interference is; the engineered-phase-reference (homodyne) detector is recovered as the complementary limit of the same expression. The mechanism is proposed to operate within ordinary quantum dynamics — a claim about the selection stage that the microscopic model owed for it has not yet supplied, and which the wave-realism premise below in any case exceeds — closes its timescale requirements in real detectors with orders of magnitude to spare, reproduces Glauber counting statistics including bunching and anti-bunching, and extends to entangled pairs as a two-stage joint game yielding the full quantum correlations with a resync-fidelity law $S = 2\sqrt{2}\,\eta$ — a construction conditional on an explicit nonlocal shared-registry premise, from which no-signaling then follows. The single-detector Born result is conditional on ordinary detector physics plus a wave-realism premise where the detector is spatially compact; where one quantum's candidate registration sites are spacelike separated — a beam-splitter, most simply — it is additionally conditional on the nonlocal, preferred-frame ontology, since exclusivity there is a constraint on an energy ledger closed across the separation. The entangled-pair extension is conditional on that ontology throughout. Unlike the axiom it replaces, the mechanism specifies its own failure conditions — concentrated in a single experimental family and testable with a single-photon tabletop discriminator — and, in the entangled sector, with a cross-station injection protocol on a working Bell setup, where broadband injection bounds the shared-registry fidelity and phase-locked injection predicts a correlation excess above the Tsirelson bound that no quantum-mechanical mechanism can produce.

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1. Introduction

Quantum mechanics rests on two kinds of law. The dynamics — the Schrödinger equation — is deterministic, linear, and continuous. The measurement postulate is none of these: it asserts that observation yields a single outcome, with probability equal to the squared amplitude, $P = |A|^2$. This — the Born rule — is where all of the probability in physics' most successful theory enters, and it is stated, not derived. It is strange twice over. First, *why probability at all*: what physical process selects one outcome when the dynamics, left to itself, selects nothing? Second, *why the square*: detectors couple linearly to fields, the dynamics is linear in amplitudes, and yet the statistics are quadratic.

A century of derivation attempts falls into two families. The *structural* family establishes uniqueness of the measure: Gleason's theorem (whose Hilbert-space-dimension-≥ 3 restriction is lifted by the POVM generalization of Busch, so the uniqueness holds even for two-outcome measurements), decision-theoretic arguments within the Everett interpretation (Deutsch; Wallace; with the circularity critique of Barnum et al.), Zurek's envariance, quantum-equilibrium typicality in Bohmian mechanics (Dürr–Goldstein–Zanghì; with Valentini's dynamical relaxation), and operational reconstructions (Masanes–Galley–Müller; and the recent controversy over their assumptions). These results say, with varying assumptions: *if* outcomes occur with probabilities respecting the structure of quantum theory, the measure can only be Born. They do not say what an outcome physically is, or what process selects it. The *dynamical* family supplies a process: stochastic state-vector reduction, from Pearle's pioneering models and the GRW/CSL program through the energy-driven and martingale formulations of Gisin, Diósi, Percival, Hughston, Brody–Hughston, and Adler–Brody–Brun–Hughston, and Stephen Adler's trace dynamics beneath them. These models produce single outcomes with Born frequencies — at two costs. They modify the fundamental dynamics, with new constants and (in principle) observable anomalies. And, decisively for our purposes, the noise that drives reduction is *engineered*: its statistical structure is chosen precisely so that the squared amplitudes evolve as a fair game (a martingale), from which Born frequencies follow; Gisin's no-signaling theorems showed that no other choice was open to them. In every dynamical model to date, the fairness that makes the Born rule exact is an input.

This paper relocates the game and argues that the fairness follows from stated detector premises rather than being engineered. The arena is not Hilbert space but the *detector*. An incident quantum excites every candidate absorber site in proportion to the local intensity — a *reversible*, distributed excitation, still ordinary wave mechanics, that commits to nothing (§3) — and the deposited energy is $e_i \propto |A_i|^2$, by nothing deeper than the quadratic energetics of driven oscillators, and this is the only place the square enters. The deposited energies then compete: a noisy, substrate-mediated exchange redistributes the quantum among sites until one site registers it. If the shares of the total evolve as a martingale, a classical theorem — Pearle's gambler's ruin, in its mature optional-stopping form — makes the registration probabilities equal the initial energy shares, which by P1 are the Born weights — given the premises, and given the wave-realism (P0) that makes the shares real stakes. That much is the dynamical-reduction engine transplanted into detector physics. The contribution of this paper is what no predecessor could supply: proofs that the game *is* fair, from premises about detectors rather than stipulations about noise. Amplitude-linear coupling forces the unique noise scaling with drift-free shares (Theorem 1); below the registration threshold an absorber has no autonomous phase and hence no lever for rich-get-richer feedback, with level discreteness enforcing the threshold's sharpness (Theorem 2); the locality of detector noise protects site independence, with the penalty for correlation computed exactly (Theorem 3); and energy conservation makes registration rates kinematically linear in occupation, rendering the statistics independent of registration speed (Theorems 4–5) — a robustness result for the fast-commit regime, since in real detectors commitment is slow and first passage settles the outcome first (§5.4, §6.1). Nothing in the construction modifies quantum dynamics; the noise is ordinary vacuum and environment noise — physically, the interference of each site's amplitude with the uncontrolled-phase fields at the detector surface (§4). Earlier drafts added here that "the Born weight is nowhere inserted by hand." That claim is withdrawn as stated. What is true is narrower and should be read exactly: no *outcome* is drawn from $|A_i|^2$, and no parameter of the selection dynamics is tuned against the comparator. But the identification of the capture energies with squared local amplitudes (P1), together with the wave realism that makes them simultaneously possessed (P0), is an explicit premise placed at the head of the argument — and it is the one place a Born weight enters the construction. The honest statement is not that nothing was inserted, but that everything inserted is declared, is confined to two premises, and is the target of §8's experiments.

Three things the argument keeps distinct, since conflating them is the natural misreading. **Settled, and**

not ours to claim: that the measure — *if* outcomes occur with quantum-structured probabilities — can only be Born is Gleason's theorem, and that a driven oscillator's deposited energy goes as amplitude-squared is classical energetics, the same $|E|^2$ that records a hologram with no probabilistic content; the square is therefore not the contested quantity, and we do not derive it. **Claimed:** a physical *selection process* — a noise-driven competition among coupled, real wave packets — that *realizes* that measure, turning deposited-energy shares into registration frequencies as a fair game whose fairness is argued from premises rather than stipulated. **Open:** deriving that process, with its conservation and covariance, from a closed microscopic model, so that identifying an energy-share with an outcome-probability (premise P1) becomes a result rather than an input (§9); and, prior to that, demonstrating pathwise that the dynamics produces *exactly one* record — exclusivity, quench, and energy routing are owed by the selection stage and are not discharged anywhere in this paper. This paper argues the middle claim, conditional on premises that make the first explicit and the third tractable; where earlier phrasings above ("derives the fairness," "nowhere inserted by hand") risk blurring the middle into the first, they should be read against this partition, and against the withdrawal recorded in the preceding paragraph.

Complex-amplitude framing. The wave should not be described as *complex-valued energy*: measured energy and the conserved energy ledger are real. What is complex is the wave amplitude, whose phase carries relational information and whose bilinear combinations — amplitude multiplied by its complex conjugate, with the appropriate field or spinor operator — define real densities and currents. The physically suggestive reading adopted here is therefore that the wave is a distributed **energy-transfer amplitude**: it carries the phase information that determines interference and the real local capacity to exchange energy with matter. A detector array then converts that distributed coupling structure into exclusive, whole-quantum events whose accumulated locations follow the corresponding quadratic weights. This observation motivates P0–P1 but does not prove their stronger ontological step. In standard quantum theory the local quadratic quantity is an expectation or detection weight, not automatically an event-by-event parcel of possessed energy; promoting it to simultaneously real stakes, and deriving why exactly one stake becomes the actual transfer, remain part of the microscopic-selection burden.

Mechanism-resolved decoherence. The aim is not to replace decoherence or to use it as a synonym for collapse, but to open the detector-side mechanism that the word often compresses. The incident field first creates a distributed field–matter excitation. Coupling to phonons, charges, disorder, and circuit degrees of freedom then disperses relative phase information into the environment and defines robust candidate material channels. That is the decoherence layer: it explains the loss of accessible interference and the emergence of a pointer basis, but not a unique event. The proposed conserving fair game is the next layer: it acts among those physically formed channels and is intended to select one actual energy-transfer route with Born frequencies. Microscopic seed commitment, detector amplification, and registration follow afterward. The paper's proposed completion is therefore a chain — **distributed capture, channel-forming decoherence, one-world selection, commitment, amplification, and record** — whose stages must be derived and tested separately rather than collapsed into a single measurement postulate.

We are explicit about provenance, because the assembled result draws on several literatures. The martingale engine is due to Pearle and to Stephen Adler, Brody, Brun and Hughston, and we claim no part of it. The proportionality of detector excitation to $|E|^2$ is semiclassical detection theory (Lamb–Scully; Mandel–Sudarshan–Wolf). Registration by a metastable apparatus tipping into one basin was analyzed in depth by Allahverdyan, Balian and Nieuwenhuizen. Threshold detection of classical random fields as a route to (approximate) Born statistics was developed by Khrennikov and, earlier, in stochastic optics. What is new here is the conjunction: the four fairness theorems and their premises; the commit-rate-independence and kinematic-linearity results that extend exactness to all registration speeds and all quantum sectors; the port-decomposition theorem governing what deviations could ever be observed; the resync-fidelity law $S = 2\sqrt{2}\,\eta$ for entangled pairs; and a falsifiable deviation package with a tabletop discriminator.

The claim is deliberately bounded. Everything is conditional on a wave-realism premise (P0, §2) and six further premises: four are standard detector physics plus one structural assumption already implicit in perturbation theory (P1–P4; these with P0 carry the single-detector theory for a *spatially compact* detector), and two are ontological commitments consumed by the multi-quantum and entangled-pair constructions, and — as v0.7 records in §2 — also by any single-quantum geometry whose candidate registration sites are spacelike separated, the beam-splitter included (P5–P6, stated explicitly so that nothing is smuggled). We do not derive probability from a probability-free theory — no one can; we do not derive quantum mechanics; we do not derive entanglement; and we do not derive the Born measure, which enters as the comparator. Earlier drafts claimed that *given the premises, the measurement postulate is redundant.* That is too strong and is replaced by the conditional it was reaching for: *given the premises, and given a frozen preparation measure over the detector's ready state, the mechanism's outcome law is compatible with the Born comparator over the domain tested* — definite outcomes, Born weights, Glauber counting statistics, and Bell correlations following from dynamics plus noise. Compatibility over a tested domain is not redundancy of the postulate: it leaves the measure itself adopted rather than explained, and it leaves the production of exactly one record owed by §5's dynamics. What the mechanism does add, and a postulate does not, is that it specifies the conditions under which its statistics would fail, concentrated in one experimental family and testable on a benchtop (§8).

The paper proceeds as follows. §2 states the premises. §3 and §4 develop capture and the selection game, including the instructive failures that make fairness a knife-edge. §5 proves the four fairness theorems. §6 verifies the timescale window in a real detector and extends the mechanism to multi-quantum states, recovering Glauber statistics and ρ-linearity. §7 treats entangled pairs: the two-stage joint game, the fidelity law, the resync mechanism, and no-signaling as a theorem. §8 assembles the deviation ledger, the protective port-decomposition, the tabletop discriminator, and the fork the mechanism cannot yet resolve. §9 situates the result and lists open problems.

2. Premises

Our claim has the form of a conditional: *given four physical premises about detectors and a frozen preparation measure over the detector's ready state, the mechanism's outcome law is compatible with the Born comparator over the domain tested* — definite outcomes, Born weights, photodetection counting statistics, and Bell correlations following from dynamics plus noise. This section states the premises. Each is a statement about the physics of detection, and none mentions outcome probabilities. Note carefully what this does *not* say: P1 assigns the capture energies the value $\kappa|A_i|^2$, and P0 makes those energies simultaneously possessed. Whether local quadratic quantities of standard quantum theory may be read as event-level possessed energies rather than as expectations or transition weights is exactly what P0 asserts and what standard theory declines to grant. $|A|^2$ therefore does carry the outcome weights into the argument — openly, as a premise, in one place. The premises are the interface: everything that follows can be evaluated as a conditional result on P1–P6 alone, independently of any particular ontology supplying them (one such program is noted in §9).

Throughout, a measurement event is decomposed into three stages — **capture** (the incident wave reversibly excites the detector), **selection** (one site ends up with the quantum), and **registration** (the outcome becomes irreversible and macroscopic). Irreversibility enters only at the last stage; that capture is reversible is an experimental fact, not a modeling choice (§3). The premises attach to specific stages; the theorems of §5 govern the selection stage, which is where the Born weights are set.

Notation. An incident single quantum of frequency ω (energy $E_0 = \hbar\omega$) illuminates a detector containing N candidate absorber sites (bound electrons or atoms), indexed i. The local wave amplitude at site i is A_i. During selection, site i holds excitation energy $e_i(t) \geq 0$; we write $E = \sum_j e_j$ for the total and $s_i = e_i/E$ for the *shares*. Outcome probabilities are denoted P_i.

P0 (Wave realism). *The incident quantum is a real, spatially distributed field configuration — not a probability amplitude over mutually exclusive absorbers. Consequently the capture energies $e_i(0)$ are simultaneously, physically present at every illuminated site.* We state this explicitly because it is the ontological commitment on which the identification of an energy-share with an outcome-probability rests. It is not standard-textbook quantum mechanics (which holds the pre-registration state indefinite); it is the realist premise of the substrate program of §9.3, and every result below is conditional on it. P0 is what distinguishes the present mechanism from a mere re-labeling of $|A|^2$: given P0, the $e_i(0)$ are real stakes and the selection dynamics is a genuine physical process, not a bookkeeping identity.

P1 (Energetics of capture). *The incident wave couples to every site within its bandwidth, driving each* *as a damped oscillator; the energy deposited at site i during capture is proportional to the squared local amplitude,*

$$e_i(0) \propto |A_i|^2.$$

This is elementary driven-oscillator energetics — the energy of an oscillator is quadratic in its displacement — and it is where the squaring enters the theory: as mechanics, not as a probability rule. The same fact appears as the small-tip-angle excitation law in magnetic resonance ($\propto \sin^2(\theta/2) \approx \theta^2/4$) and as the semiclassical photodetection rate $\propto |E(\mathbf{r})|^2$ of Lamb and Scully [Lamb & Scully 1969] and Mandel–Sudarshan–Wolf [Mandel et al. 1964]. We emphasize a discipline that P1 imposes: at the capture stage *every* illuminated site is excited, in proportion to the local intensity. The single-site character of the eventual event is the *output* of selection, never an input. (Reducing the perturbative sum over sites to a single absorption diagram already presupposes the Born rule; we do not do so.)

P2 (Amplitude-linear coupling). *All couplings in the selection dynamics — the drive, site–field exchange, and the vacuum/environment noise — act linearly on wave amplitudes.* Consequently a noise force $\xi(t)$ does work on site i at a rate proportional to its amplitude $\sqrt{e_i}$, so the stochastic increment of the site energy has the form

$$de_i = (\text{drift terms}) + \sigma\sqrt{e_i}\, dW_i,$$

with energy-noise *variance* proportional to e_i. P2 is not introduced for the present purpose: it is the same structural assumption that underlies golden-rule transition rates throughout perturbation theory — couplings act on amplitudes, and rates emerge quadratic. The burden of this paper is to show that this familiar assumption *also* fixes the outcome probabilities; the sensitivity of that result to P2 — and the failure of every alternative noise scaling — is quantified in Theorem 1. (A physical origin for exactly this scaling — the interference cross term between a site's complex amplitude and the uncontrolled-phase fields at the surface — is developed in §4.)

P3 (Detector initial conditions). *(a) The internal phases of the absorber sites are initially mutually incoherent: no phase relationship exists among sites, or between any site and the incident wave, before capture. (b) The noise driving the selection dynamics is dominantly local: writing the site-noise correlation matrix as C_{ij}, the spatially correlated (radiative) fraction of the noise power is small, $f = \Gamma_{\text{rad}}/\Gamma_{\text{tot}} \ll 1$, so that $C \approx \mathbb{1}$ at working accuracy.* Both clauses are checkable statements about real detectors. (a) holds for any absorber ensemble in internal equilibrium — mutual coherence among detector electrons is not a natural state and would have to be engineered. (b) holds because the linewidths of solid-state absorbers are dominated by local scattering (phonon, carrier–carrier; correlation length ℓ of nanometres; $\Gamma_{\text{loc}} \sim 10^{13}$–$10^{14}\,\text{s}^{-1}$) while the wavelength-scale-correlated radiative channel contributes only $\Gamma_{\text{rad}} \sim 10^8\,\text{s}^{-1}$: $f \sim 10^{-6}$.

Correction to the stated scaling (2026-08-31). Earlier drafts of this premise wrote the residual off-diagonal correlation as $C = \mathbb{1} + O(f)$. That is the f-dominated branch only, and it is *not* the binding one at the value of f this paper quotes. Under the fluctuation–dissipation theorem the noise covariance is not an independent object: it is fixed by the damping matrix Γ, whose off-diagonal structure carries both the wavelength-correlated radiative part and the exponential tail of local scattering. The correct statement is

$$C = \mathbb{1} + O\big(\max(f,\ e^{-d/\ell})\big),$$

with d the spacing between absorber sites. Numerically (six-site check, `code/check_g1g2_ready_state.py`): at $f = 10^{-1}$ and 10^{-3} the residual tracks f exactly, but at $f = 10^{-6}$ with $d = 20\,\text{nm}$ and $\ell = 2\,\text{nm}$ the residual is 4.6×10^{-5} — some $46\times$ larger than f, because the nearest-neighbour term $e^{-d/\ell}$ dominates. Two consequences. First, the correlation floor in a real detector is set by the *local* correlation tail, not by the radiative fraction, so quoting $f \sim 10^{-6}$ understates it. Second, and more consequentially, the premise now carries an explicit geometric condition, $d \gg \ell$: in a densely packed absorber array with $d \lesssim \ell$ the nearest-neighbour noise correlation is $O(1)$ and clause (b) simply fails. **This condition should be checked for any detector to which the mechanism is applied and is not yet verified for the detector families discussed in §6.1 and §8.** It does not weaken the premise where it holds — both terms remain tiny — but the scaling as previously stated was wrong in exactly the regime claimed.

Theorem 3 computes the exact penalty when either clause fails — P3 is where the mechanism is *falsifiable*, and §8 [to draft] builds its experimental proposals from precisely these failure modes. Both clauses of P3 are, in the microstate contract of the companion note, consequences of the detector's ready-state measure rather than independent premises — (a) from the $U(1)$ invariance of that measure, (b) from the fluctuation–dissipation relation above; that contract is drafted but not independently reviewed, so P3 is retained here as a premise.

P4 (Registration taxonomy, and the commitment threshold). *(a) Registration consumes the full quantum, in both absorber classes: an outcome is written only when $\hbar\omega$ is delivered at one site. (b) Registration is of one of two kinds, according to the absorber's spectrum. (i) Discrete-level absorbers (atoms, molecules): no stationary state exists below the transition energy, so the threshold in (a) is enforced spectroscopically. (ii) Continuum absorbers (semiconductors, metals): final states exist at all energies above the gap, so the threshold is enforced by the energy ledger rather than by the level structure, and a commitment channel operates alongside selection.*

Clause (a) is new to v0.7 and is stated as a premise because the exclusivity argument depends on it. Under a closed pot $\sum_i e_i = \hbar\omega$, a full-quantum threshold means at most one site can ever be above threshold — two would require more than the available energy. Exclusivity is therefore a consequence of energy conservation, not an imposed stopping rule, and nothing in the construction samples an outcome index. Earlier drafts left (a) implicit in Theorem 5's energy accounting; it does too much work to remain implicit. One qualification, added in v0.8: clause (a) is a premise about the *quantum*, not a consequence of the absorber's level structure. Wherever $E_{\text{photon}} > 2E_{\text{gap}}$ — silicon below 553 nm, and every superconducting absorber — the closed pot could pay for two real sub-gap excitations, and only (a) forbids it. The gap enforces the full-quantum threshold on its own only where $E_{\text{photon}} < 2E_{\text{gap}}$ (§6.1).

Clause (b)(ii) was weakened in v0.7, which replaced earlier drafts' "a rate process operating throughout selection" with the claim that, in the regime §6.1 then established, both classes reach an outcome by *first passage* (Theorem 0) long before any commitment channel fires. v0.8 withdraws that in turn. The v0.7 reading rested on §6.1's $\tau_{\text{commit}} \sim$ ns–μs, which is the record-writing cycle, not commitment as this paper defines it; placed at first irreversibility (§6.1, v0.8), commitment in a continuum absorber is available to any site whose share exceeds $\theta = E_{\text{gap}}/E_{\text{photon}}$, and the separation between the leader's exposed climb and that clock is marginal for silicon and inverted for superconducting absorbers. So the original phrasing was closer to right than v0.7 allowed: a commitment channel *does* operate alongside selection — gated at θ, and open for a window comparable to the commit time. Theorems 4–5 are accordingly retained as load-bearing for class (ii): Born survives there to the extent the commit law is linear in the site's holding (Theorem 5), and the linearity required is that of the first-irreversibility vertex — see §5.4.

v0.9 goes one step further, and records it as a choice (header). Under reading B a site's holding below the gap is an amplitude, not an energy that lacks a final state, so there is no gate at θ: the commitment channel is open from every site, at the golden-rule rate, linear in the holding. Theorems 4–5 then carry class (ii) at any commit speed, with nothing left for the ladder's top rung to protect. The θ-gated statement of the preceding paragraph is retained as reading A's version, quantified and bounded in §6.1.

The final two premises are of a different character: they are the ontological commitments consumed by the multi-quantum and entangled-pair constructions (§6.2–6.3 and §7) — and, as the third remark below records, by any single-quantum geometry whose candidate registration sites are spacelike separated. They are stated here so that the interface is complete and nothing is smuggled in later.

P5 (Shared registry). *Whenever a state's support spans candidate registration sites that are not causally connected on the timescale of the selection game, the shared amplitude over those sites — the single-quantum amplitude $A(\mathbf{x})$ delocalized across a beam-splitter's arms; the pair registry $\psi(i,j)$ of a two-quantum state; the joint coefficients c_{ij} of an entangled pair — is physically instantiated in the medium connecting the regions, and a registration event renormalizes it faithfully: the surviving shared amplitude re-forms around the realized outcome, so that the weights delivered to the remaining regions are exact (update fidelity $\eta = 1$).*

The scope of P5 is broadened in v0.7. Earlier drafts stated it for "multi-quantum and multi-partite states" only, which left the paper's own beam-splitter argument (§3) without a premise: one photon delocalized over two arms is neither multi-quantum nor multi-partite, yet §6.1's ps-scale game cannot span ns-scale arm separations, so the energy ledger must still be closed across them (§2, third remark). The broadened statement covers that case as its simplest instance and the entangled pair as its richest. P5 is the locus of the mechanism's nonlocality, held explicitly rather than implicitly.

The two sectors are not on the same footing, and the difference is a result rather than a stipulation. **In the single-quantum sector $\eta = 1$ is forced, not measured.** Suppose registration completes at a site in arm A, consuming the full quantum (P4(a)), while a fraction $(1-\eta)$ of arm B's share survives the update. The ledger then holds $\hbar\omega\,[1 + (1-\eta)/2]$ for a balanced splitter — energy created from nothing. Since the pot is closed by construction, $\eta < 1$ is not a degraded regime but an inconsistent one, and exact anticorrelation follows with no free parameter. In the entangled sector, by contrast, two quanta are separately available, $\eta < 1$ misallocates conditional weights without violating conservation, and η is a genuine measurable — §7.2 shows it is experimentally $\gtrsim 0.99$, and §7.3 argues geometrically why it should be exactly one. The single-quantum argument now supplies an independent reason for the same conclusion: if the substrate's update mechanism does not distinguish the two sectors, conservation pins η in one of them and the other inherits it.

P6 (Ordering foliation). *Registration events are ordered by a global foliation (a preferred time-slicing); for spacelike-separated commitments, "first" refers to this ordering.* P6 is consumed in §7, to make the two-stage construction well-defined, and also wherever spacelike-separated candidate sites compete for one quantum (§2, third remark) — not, as earlier drafts had it, in §7 alone; §7.1 shows the outcome statistics are independent of the ordering, and §7.3 discusses the experimental status of preferred-frame effects.

Three remarks on the status of the premises. First, they are logically independent handles: each theorem of §5 consumes specific premises, and each experimental proposal of §8 targets a specific premise's failure. Second, P1–P4 are not exotic: P1 and P4 are standard detector physics, P3 is standard detector *matter* physics, and P2 is the structural assumption already underlying the perturbative rate laws of ordinary quantum optics.

A qualification on that second remark, which earlier drafts did not make and which a reader should apply throughout. Calling these premises "standard detector physics" describes their *lineage*, not their epistemic status in this argument. The deposited-energy law $e_i(0) = \kappa|A_i|^2$ and the commitment-rate structure of P4(ii) are **prior components of this model**, adopted here and carrying the model's assumptions with them; they are not independently established microscopic facts about event-level energy in a single-quantum absorption. Standard detection theory licenses $\propto |E(\mathbf{r})|^2$ as an *ensemble rate* or an expectation. Reading the same expression as a set of simultaneously possessed event-level energy shares is P0's work, not the literature's, and the literature does not grant it. Wherever the text below says a premise is "standard," read: standard as a rate law, promoted to an event-level ontological claim by P0. Third, P5 and P6 are where the premises exceed standard physics. Earlier drafts added that they are *quarantined* — that no single-detector result (§§3–5, §6.1) uses either. **That claim is withdrawn in v0.7; it is false, and the error was ours rather than a reviewer's.**

The counterexample is inside this paper. §3 assigns Stage 2 the job of supplying exclusivity for a single quantum split between two *separated* detectors — beam-splitter anticorrelation, the observation that a single photon never yields a coincidence [Grangier et al. 1986]. Under the mechanism, exclusivity is energy conservation over a closed pot with a full-quantum threshold (P4(a)). For the beam-splitter that pot must be closed *across the two arms*. Two consequences follow, and neither is optional:

- **The arms cannot be causally connected during the game.** §6.1 puts τ_{game} at 12–81 ps, while arms a metre apart are 3.3 ns apart in light travel time. If each arm's sites competed only among themselves, a balanced beam-splitter would leave each arm holding $\hbar\omega/2$, neither could ever reach the full-quantum threshold, and *neither would ever register*. The observed anticorrelation therefore requires the pot to be shared across the separation — a superluminal constraint on the energy ledger, not a local one.
- **Ordering is required for spacelike commitments**, which is exactly P6's content.

So the single-detector sector consumes a nonlocality premise whenever the incident quantum's support spans spacelike-separated registration sites. Worse, the premises as they stood did not cover this case: P5 was stated for "multi-quantum and multi-partite states," and a single photon delocalized over two arms is neither. **P5's scope is accordingly broadened in v0.7** to any state whose support spans candidate registration sites that are causally disconnected on the game's timescale, with P6 supplying the ordering; §§3–5 are no longer advertised as resting on P0–P4 alone. The broadening was done in one pass with §7, and it turns out to cost less than feared: in the single-quantum sector the update fidelity is *forced* to unity by energy conservation rather than added as a parameter (P5; §7.2).

Two calibrations, so the cost is neither hidden nor overstated. This is a real increase in the price of the single-detector theory: the claim that the cheap sector is cheap does not survive. But it is not a defect peculiar to this mechanism. Any single-world account of beam-splitter anticorrelation carries a nonlocal constraint — textbook collapse is nonlocal too — and what changes here is only that the framework must own the premise explicitly rather than inheriting it silently from the measurement postulate. Any ontology supplying P5–P6 supports the constructions built on them; one concrete realization — a substrate framework in active, publicly documented development — supplies all six premises and motivated this investigation [Framework repository; see §9].

3. Stage 1 — capture

When the incident quantum reaches the detector, its wave extends over the whole illuminated patch, and every candidate site within its bandwidth responds. Each site is a bound charge with a transition frequency near ω; the wave drives each as a damped oscillator, inducing a polarization of amplitude proportional to the *local* field amplitude A_i, with the wave's phase imprinted on the response. The situation is precisely that of an RF pulse in magnetic resonance: the pulse tips every spin in the excited slice, each by an angle set by the local B_1 amplitude, and stamps its own phase on the resulting precession. At the end of capture the detector carries a *distributed* polarization pattern mirroring the incident intensity profile. No site has been singled out.

We flag a discipline that this picture imposes and that the paper maintains throughout. It is conventional to speak of "the photon interacting with an electron," as in the one-photon–one-vertex absorption diagram. But that diagram is one term in a coherent sum over all sites, and the step that reduces the sum to a single term — assigning the event to *this* electron with weight $|A_i|^2$ — is exactly the Born-weighted selection whose origin is in question. To assume it at capture is to beg the question the paper answers. The plural must be kept alive until the dynamics kills it; selection is the *output* of Stage 2, never a kinematic input.

The energy law (P1). For a resonant secular drive, the induced oscillation amplitude at site i is proportional to A_i, and the stored energy is quadratic in the amplitude:

$$e_i(0) = \kappa|A_i|^2,$$

with κ common to all sites (and hence cancelling from the shares s_i). The relative phase between the drive and a site's internal clock affects only the transient, not the secular tip: the drive imprints its own phase, so the deposited energy is phase-independent — a point that matters in §5.2, where the *absence* of surviving phase relationships does real work. In magnetic-resonance language, a small tip angle $\theta \propto B_1 t$ excites with probability $\sin^2(\theta/2) \approx \theta^2/4$ — quadratic in the drive. This is the only place the squared amplitude enters the theory, and it enters as oscillator mechanics. Holography is the same fact as an engineering discipline: recording media respond to the local squared field amplitude — and the fringes survive the exposure's time integration only because reference and object share a fixed phase, mutual coherence being what renders the energy distribution stationary and hence recordable [Mandel & Wolf 1995]. Computer-generated and digital holograms, in which the pattern is computed from wave theory, fabricated, and successfully reconstructed, verify the quadratic law end-to-end in a setting with no probabilistic reading. (At single-photon exposures, which grain records each quantum is the selection problem of §4; and coherence here is between the fields at capture — pre-established coherence among the absorbers themselves is the separate, testable matter of P3(a) and §8.) Holography also supplies the *control case* for a distinction developed in §4: the single coherent reference is what makes the recorded interference pattern *invertible* — the stored fringes point back to the wavefront, and reconstruction (including phase-conjugate readout, which runs the wavefront backward) is possible precisely because every grain's record refers to one shared phase standard. An ordinary detector bulk is, by P3(a), the opposite configuration — each site beats the incident wave against its own uncontrolled reference — and §4 traces what becomes of the phase record in that case.

Relation to semiclassical detection theory. Capture, so described, is the semiclassical photodetector of Lamb and Scully: a quantized absorber driven by a (classical) field reproduces the photoelectric threshold, prompt emission, and a response proportional to $|E(\mathbf{r})|^2$, with no photon concept required. We adopt that result and its credit wholesale. What semiclassical theory famously lacks is *exclusivity*: a classical pulse carrying one quantum of energy, split between two separated detectors, excites both and forbids neither from registering — whereas experimentally a single photon never produces a coincidence (anticorrelation at a beamsplitter). Supplying exclusivity — one full quantum at one site, the rest relaxed — *without importing the Born rule to do it* is precisely the job assigned to Stage 2. Note finally that energy conservation fixes the total: the captured energy sums to the one quantum, $E(0) = \hbar\omega$, which is why the normalization $\sum_i P_i = 1$ will come out automatically rather than by decree.

Capture is reversible. Nothing in Stage 1 is a commitment. The distributed polarization is a coherent excitation of ordinary wave mechanics, and it can be run backwards — not merely in principle but on the bench. The spin echo [Hahn 1950] is precisely this demonstration in the magnetic-resonance setting: excitation deposited across an ensemble, apparently lost to dephasing, is refocused and *re-emitted*, no registration ever having occurred. Its optical twin, the photon echo [Kurnit et al. 1964], shows the same reversal for optical absorbers, and atomic-frequency-comb quantum memories complete the point at the single-quantum level: a single photon is captured across a large ensemble of absorbers and later re-emitted on demand, with the storage step deliberately engineered to be selection-free [de Riedmatten et al. 2008; Afzelius et al. 2009]. Capture-without-registration is thus a working technology, and it fixes an important interpretive point: the deposition of energy across the detector face is *not* the collapse, and the capture stage sits entirely on the settled side of §1's partition — semiclassical in its response law and reversible in its dynamics. Irreversibility, and with it the arrow that makes an outcome an outcome, enters only at registration, at the bottom of the temporal ladder of §6.1. Everything contested in this paper happens in between.

4. Stage 2 — selection as a stochastic energy game

Capture leaves the detector with tip energies $e_i(0) \propto |A_i|^2$ summing to one quantum; observation finds *one* site registered with the full quantum and the rest relaxed. Stage 2 is the dynamics connecting the two, and its physical content is **exchange**: the tipped sites are coupled to the common field and, through it, to one another; vacuum and environment noise (P2, P3) drives a stochastic redistribution of the deposited energy among them. The process terminates at one of the absorbing configurations supplied by P4: a site that assembles the full quantum crosses the locking threshold and registers (class (i)), or registration fires as a rate process during the exchange (class (ii)); a site whose energy relaxes to zero drops out of the competition.

Two features of the noise deserve immediate comment. First, its role is that of a *tie-breaker*: an infinitesimal symmetry-breaking seed determines which absorbing configuration a given run reaches, but — as the theorems below make precise — the noise *magnitude* drops out of the outcome statistics entirely. The noise decides *that* symmetric alternatives get broken, not *toward whom*, and it carries no outcome information of its own; it is not a hidden variable, and the mechanism is not a local-hidden-variable model. Second, a guardrail we state now and honor in §7: in spatially extended (entangled-pair) configurations, the relevant fluctuations belong to the *shared* substrate, not to independent local reservoirs at each station — a locality caveat that Bell's theorem makes mandatory for any mechanism of this kind.

The physical origin of the noise: interference at the surface. The interaction at the detector surface is a complex-amplitude affair, and keeping it complex is what identifies the noise physically. Write the complex oscillation amplitude of site i as a_i, so $e_i = |a_i|^2$, and let the site sit not only in the incident wave but in everything else present at the surface: the wave packets of the detector's other bound charges and the vacuum/environment field, contributing over a short interval a fluctuating complex amplitude increment δb_i of unknown magnitude and — by P3(a) — uncontrolled phase. The site's energy update then contains

$$|a_i + \delta b_i|^2 = |a_i|^2 + 2\,\mathrm{Re}(a_i^*\,\delta b_i) + O(|\delta b_i|^2),$$

and the interference cross term is the noise: it has *zero mean* because the surface holds no phase reference against the site (P3(a) — averaging over the phase of δb_i kills it in expectation), and its fluctuation is linear in $|a_i| = \sqrt{e_i}$, so the energy-noise *variance is proportional to* e_i — precisely the P2 form whose uniqueness

Theorem 1 establishes. On this reading P1 and P2 are the two halves of one complex expression: the direct term $|a_i|^2$ is the capture law, and the cross term with the uncontrolled-phase surroundings is the fair noise. Note what the argument does *not* require: the magnitudes, spectra, or detailed wave packets of the surrounding charges and vacuum modes never enter — the noise magnitude drops out of the outcome statistics entirely — only linearity (P2) and phase-randomness (P3(a)) are consumed, and both are premises already in place. The unknowability of the surface fields is thus load-bearing rather than regrettable: the martingale theorem makes every property of the noise beyond its first two moments provably irrelevant, so fairness is a *universality* statement — the odds depend only on a symmetry (no phase reference at the surface), the way diffusion depends on molecular chaos and not on molecular trajectories — and the one dial engineering can turn, the fluctuation amplitude, sets rates, never odds. The fairness of the game is thus the statement that *nothing at the surface holds a phase reference against the players*; correspondingly, the engineered violation of P3(a) is a known instrument, not a pathology — beat the signal against a strong coherent reference wave (a *local oscillator*, the homodyne detector of quantum optics) and the same cross term stops averaging away and becomes a signal *linear* in the complex amplitude, which is exactly how homodyne detection behaves [Mandel & Wolf 1995]. One complex expression, two limits: uncontrolled surface phases give intensity capture plus fair noise; an engineered phase reference gives linear field readout. We present this as a mechanism sketch — a scaling argument grounding P2's form, not the closed microscopic derivation with conservation and drift terms tracked, which remains open (§9.4); the panel's identification of that derivation as the central open problem stands.

The companion terms and the fairness window. The same expansion that produces the cross term produces two companions, and the sketch is honest only if it tracks them. The bath's own term $|\delta b_i|^2$ has a mean — heating, and with it the dark counts of a warm detector, which are the selection game won by the bath with no incident quantum — and a fluctuating part whose variance is *independent* of e_i: an additive noise floor. Theorem 1 computes exactly what an additive component does to the game: it drifts shares toward the leaders. Fluctuation–dissipation (the theorem locking the strength of the drag a system feels to the strength of the jiggle it receives) likewise obliges the damping $-2\gamma e_i\,dt$ to accompany the noise; being linear in e_i, it is share-neutral for site-uniform γ (Appendix B.3). The mechanism's own logic therefore imposes an operating condition: Born statistics are clean where the stake dwarfs anything the bath can play with — site energies far above the bath's own scale, which for a thermal environment is the condition $\hbar\omega \gg k_B T$. This is not an embarrassment but a consistency check against the entire detector industry: at 300 K, $\hbar\omega/k_B T \approx 96$ for a 500-nm photon, and optical photon counters work at room temperature; ≈ 5 at 10 μm, and mid-infrared detectors must be cooled; $\ll 1$ at microwave frequencies, and microwave photon counting requires millikelvin cryostats. The window in which the game is provably fair is the window in which quantum detectors are, in fact, operated — and detector cooling, on this reading, protects the statistics themselves, not merely the dark-count rate. The boundary of the window is a candidate signature (§8.6(v)); the closed calculation tracking all three terms in a conserving, quantized model is the open problem of §9.4.

History bookkeeping: the diffused hologram. [Added in v0.5.3; sponsor-originated interpretation, labeled as such — an organizing reading of the mechanics above, not an additional derivation.] The cross-term picture assigns a fate to the incident wave's *phase history.* In a hologram (§3), one shared reference makes every grain's interference record mutually registered: the ensemble record is invertible, the stored history points back to the wavefront, and capture can be run in reverse. In a detector bulk satisfying P3(a), each site — and each subsequent hop of the exchange — beats the wave against its *own* uncontrolled reference: every site writes a perfectly good local interferogram relative to a phase standard nobody recorded. Such a *diffused hologram* cannot be inverted; no operation can reassemble the wavefront from it. Three consequences. (i) The same premise does double duty: P3(a)'s reference incoherence is at once why the noise is unbiased (the fair game) and why the record is non-invertible (commitment is possible) — fairness and irreversibility have a common origin. (ii) The phase information is not destroyed but *converted*: the cross terms that would have been recordable fringes under a common reference are exactly the zero-mean fluctuations that drive the selection game — the wave's own diffused history is the tie-breaker. (iii) The slaved phase of Theorem 2 is, in this language, a *memory*: a sub-threshold site's phase points back to the drive, which is why κ_{ret}-return is possible at all; an autonomous phase has forgotten the drive, which is why only above-threshold sites can commit. Reversibility below the layer and commitment above it are the same fact about memory, seen from both sides. The echo and quantum-memory systems of §3 are then *engineered history preservation* — a common or deterministically known reference structure imposed on purpose — and the reversible/irreversible dephasing split of magnetic resonance is the same trichotomy: common reference (invertible), known-different references (invertible by rephasing), random references (non-invertible, the P3(a) case).

The engine that converts this setup into probabilities is classical, and it is not ours:

Theorem 0 (Engine; Pearle 1976, 1982; Adler–Brody–Brun–Hughston 2001). *Let the shares $s_i(t)$ evolve as a martingale (zero drift: $\mathbb{E}[ds_i \mid \mathcal{F}_t] = 0$) with absorbing configurations at $s_i = 1$ (site i registered) and suppose selection terminates with probability one. Then*

$$P_i = s_i(0) = \frac{e_i(0)}{E(0)} \overset{\text{P1}}{=} \frac{|A_i|^2}{\sum_j |A_j|^2} \,,$$

exactly, for any N and any initial distribution. The proof is the optional stopping theorem: s_i is a bounded martingale, so its expectation at the (almost-surely finite) stopping time equals its initial value; at stopping, $s_i \in \{0,1\}$, so that expectation *is* P_i. In gambling language: a fair player's expected fortune is constant, so the probability of eventually holding the whole pot equals the fraction of the chips held at the start. This engine is entirely due to the dynamical-reduction literature — Pearle's gambler's-ruin argument and its mature martingale formulation — and we claim no part of it. What that literature postulates, and what §5 derives, is the *fairness*: the zero-drift property, together with its protection against the physical effects that would spoil it.

Fairness is a knife-edge. That the fairness cannot simply be waved through is best shown by exhibiting how decisively natural-looking alternatives fail (simulation details Appendix D).

(a) Selection by threshold crossing. Suppose the winner is the site whose instantaneous drive overlap is largest, $\arg\max_i A_i|\cos\theta_i|$, with θ_i the random site phases — an extreme-value rule rather than a first-passage one. For two sites this reproduces Born deceptively well (0.791 versus 0.800 for amplitude ratio 2:1 — an accident of arcsine statistics), which is why the picture survives casual testing; the threshold detectors of stochastic optics are this construction carried through in detail [Marshall & Santos 1988]. But on the ten-site test configuration the bright site wins with probability 0.79 against a Born weight of 0.50. The structural reason is that threshold crossing is governed by an *exceedance* probability: for a site whose fluctuation variance scales with its deposited energy, $\sigma_i^2 \propto e_i \propto |A_i|^2$, the chance of crossing a fixed level a goes as $\exp(-a^2/2\sigma_i^2)$, so the odds between two sites are exponential in $1/|A_i|^2$ — not proportional to $|A_i|^2$. Extreme-value selection over many draws is dominated by the largest scale, the same disease that makes laser mode competition nearly deterministic, and the discrepancy *grows* with the number of illuminated sites — precisely the wrong direction for a rule that must hold for a detector face of any size. A threshold model that does land on Born has been tuned into a linear-response window by hand, which is the engineering of the noise that §5 exists to avoid.

The diagnosis worth carrying forward is that the rare large excursion has its *causal role* inverted. Under the fair game a winning site does reach the shell by a rare large excursion; that is what a first passage looks like. The difference is that there the excursion is the *endpoint* of an unbiased walk whose odds were set by where it began ($s_i(0) = |A_i|^2/\sum_j |A_j|^2$, Theorem 0), whereas here it is the *selection criterion* and the odds are set by the tail of a distribution. Same picture, opposite direction of causation, and only one of the two is quadratic in the amplitude. Two further commitments of the threshold picture break the construction independently of the statistics: partial absorption — the site takes "the energy above threshold" — has no referent under P4(i), since a discrete-level absorber has no stationary state below the transition for a partial deposit to occupy; and if the sub-threshold remainder departs as a free wave, the pot is no longer closed, energy that leaves irreversibly cannot be a fair stake (§5.2), and a surviving residual is in any case a coincidence generator of the kind beam-splitter anticorrelation excludes [Grangier et al. 1986]. Outcome selection cannot be a snapshot comparison.

(b) Drift-contaminated exchange. Suppose instead a prolonged exchange of fixed-size energy parcels, clipped so no site goes negative. The clipping makes the zero boundary reflecting — a depleted site can only gain — which is an upward drift for small stakes; the simulated outcome distribution on a three-site configuration is $(0.46, 0.27, 0.26)$ against Born's $(0.64, 0.29, 0.07)$. A rule that *looks* symmetric fails completely if the

martingale property fails microscopically. A stakes-scaled exchange ($\delta \propto \pm \min(e_i, e_j)$, symmetric sign) — the discretization that does satisfy zero drift — reproduces Born within the discretization error ($\sim 2\%$ at coarse step, exact in the continuum limit by Theorem 0).

These failures define the burden of §5 exactly: the physics must place the exchange precisely on the fair point (Theorem 1), protect it against self-amplifying leaders (Theorem 2) and against noise correlations among sites (Theorem 3), and terminate it without distortion (Theorems 4–5). None of these may be assumed; all must come from the premises.

5. The fairness theorems

Every dynamical account of outcome selection before this one obtains Born statistics by *engineering* the noise so that the engine theorem applies; Gisin's no-signaling results [Gisin 1989; Gisin 1990] explain why they had no choice. The content of this paper is that no engineering is needed: the fairness is forced by the premises of §2. We prove this as four theorems, each answering one physical objection to fairness. The objections are not strawmen — §4's simulations show that plausible alternatives (extreme-value selection; state-independent noise; correlated noise; nonlinear registration) each destroy Born statistics decisively. Fairness is a knife-edge, and the physics must place the system exactly on it. It does.

5.1 Theorem 1: the noise cannot cheat (fairness from P2)

Theorem 1 (Fair noise). *Under P2 and P3(b) (independent site noise), with $de_i = \sigma\sqrt{e_i}\, dW_i$ and independent Wiener increments dW_i, every share $s_i = e_i/E$ is a martingale:*

$$\mathbb{E}[ds_i] = 0 \quad \text{identically, for all } N \text{ and all configurations.}$$

Proof. Itô's lemma applied to $s_i = e_i / \sum_j e_j$ gives $\mathbb{E}[ds_i] = \frac{1}{2}\sum_j \sigma^2 e_j\, \partial^2 s_i/\partial e_j^2\, dt$. With $\partial^2 s_i/\partial e_j^2 = -2\delta_{ij}/E^2 + 2e_i/E^3$,

$$\sum_j e_j \frac{\partial^2 s_i}{\partial e_j^2} = -\frac{2e_i}{E^2} + \frac{2e_i E}{E^3} = 0. \qquad \blacksquare$$

The force of the theorem lies in its converse: the $\sqrt{e}$ scaling is the *unique* fair point. For state-independent (additive) noise, the same computation yields $\mathbb{E}[ds_i] = (\sigma^2/E^2)(Ns_i - 1)\, dt$ — a drift *toward the leaders* (rich-get-richer); for multiplicative noise (variance $\propto e_i^2$), $\mathbb{E}[ds_i] = (\sigma^2 e_i/E^2)(\frac{1}{E}\sum_j e_j^2 - e_i)\, dt$ — a drift *toward equality*. Both destroy Born statistics, in opposite directions, and simulation confirms the sign and magnitude of each failure: on a ten-site test configuration (nine sites of unit amplitude, one of triple amplitude; Born weight of the bright site 0.500), the $\sqrt{e}$ law reproduces Born at the absorbing boundary (to the Monte-Carlo floor), additive noise drives the bright site toward certainty, and multiplicative noise crushes it to 0.32 (Fig. 2; simulation details Appendix D). (The corrected suite of Appendix D also shows that stopping at a *finite* share rather than at the true absorbing boundary imparts a computable $O(1-\text{threshold})$ bias — e.g. +2.6% on the two-site 2:1 configuration at threshold 0.95 — which vanishes as the boundary $\to 1$; the physical boundary is unity because synchronization completes the lock, §6.1.) Amplitude-linear coupling is thus precisely the knife-edge — and it was in the theory before this problem was posed, doing the independent work of generating golden-rule rates. One postulate, both jobs; we regard this coincidence of roles as the strongest internal evidence for the mechanism.

5.2 Theorem 2: the rich cannot cheat (no autonomous phase below threshold)

The generic disease of winner-take-all dynamics is positive feedback: a leader that couples more strongly *because* it is leading wins with probability approaching one, not with its Born weight (laser mode competition is the canonical example). The bias operates through phase: to extract energy from the common field systematically, a site must *hold a phase relationship* with it. Theorem 2 shows that a sub-threshold site cannot.

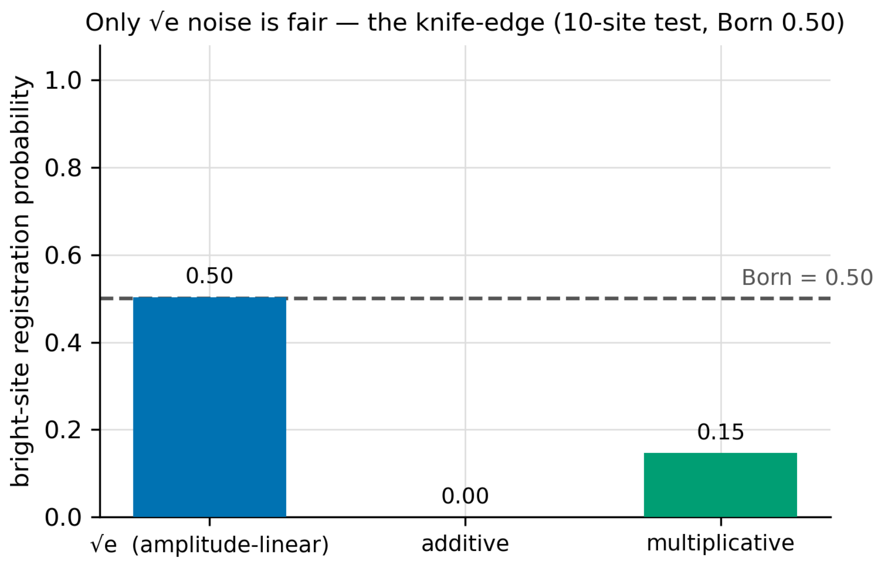


Figure 1: **Figure 2.** The knife-edge (10-site test configuration, Born weight of the bright site 0.50). Only amplitude-linear ($\sqrt{e}$) noise is fair; state-independent (additive) noise drives the bright site toward certainty, and multiplicative noise over-equalizes. Corrected absorbing-boundary simulation.

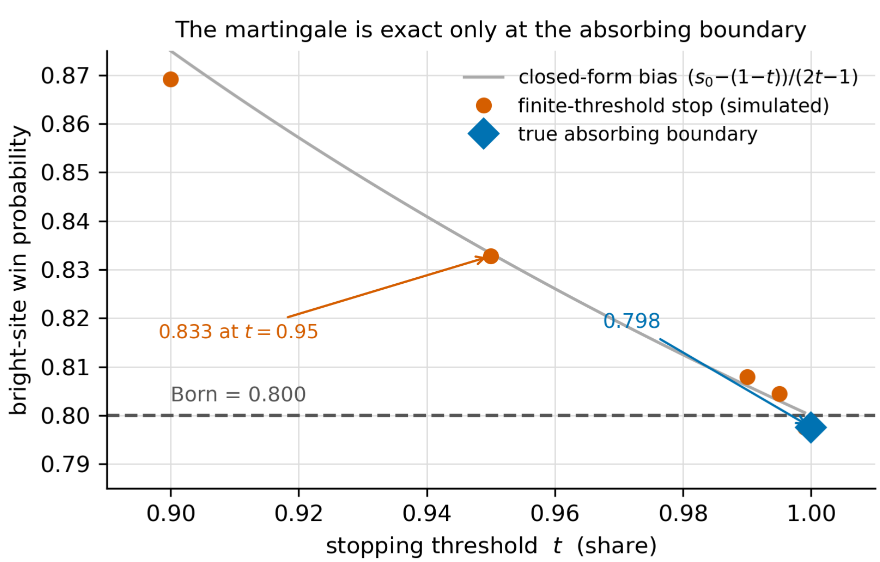


Figure 2: **Figure 3.** The martingale reproduces Born only at the true absorbing boundary. Stopping the game at a finite share and taking the argmax biases the winner probability upward by the closed-form amount $(s_0 - (1-t))/(2t-1)$ (grey curve); the two-site 2:1 configuration gives 0.833 at threshold $t = 0.95$, converging to Born (0.80) as $t \to 1$, and the true absorbing boundary (last survivor) reproduces Born to the Monte-Carlo floor. This is the artifact identified in review and its resolution.

Theorem 2 (Slaved phase). *Under P1 and P4(i), a site holding energy $e < E_0$ is off-shell by $\Delta E = E_0 - e$ and constitutes a driven linear mode that returns its excitation to the common field at the rate $\kappa_{\text{ret}} = \Delta E/\hbar$. In the rotating frame, with amplitude $a = re^{i\varphi}$, detuning Δ, and drive $fe^{i\varphi_d}$:*

$$\dot{r} = -\kappa_{\text{ret}}\, r + f\cos(\varphi_d - \varphi), \qquad \dot{\varphi} = \Delta + \frac{f}{r}\sin(\varphi_d - \varphi).$$

In Cartesian coordinates this system is linear with eigenvalues $\lambda = -\kappa_{\text{ret}} \pm i\Delta$; it possesses a unique, globally attracting fixed point $r^* = f/\sqrt{\kappa_{\text{ret}}^2 + \Delta^2}$, $\varphi^* = \varphi_d - \arctan(\Delta/\kappa_{\text{ret}})$. The phase is an enslaved coordinate: perturbations decay at rate κ_{ret}, and no winding (phase-slip) solutions exist.

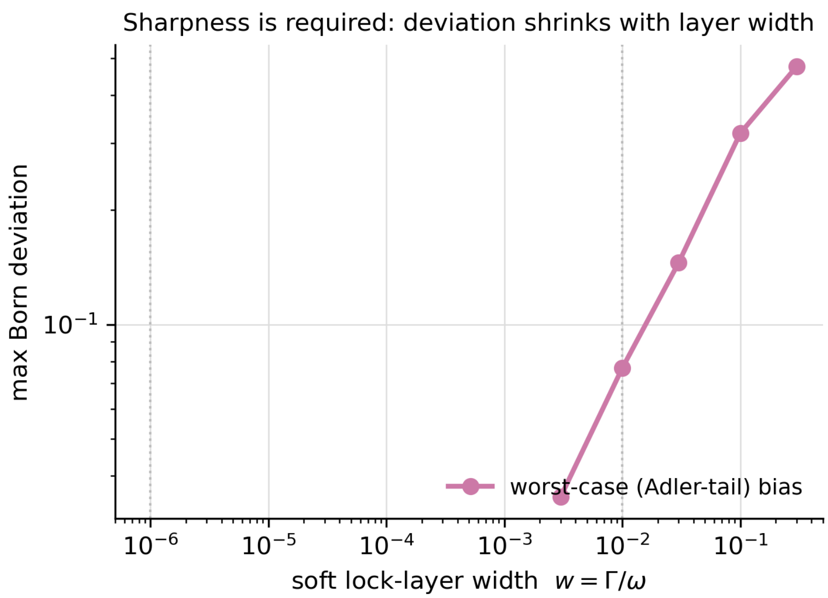
What kind of rate κ_{ret} is — and is not. [Added in v0.5.3, prompted by finding N2 of the companion measurement-structure paper's review panel; clarification articulated by the sponsor.] κ_{ret} is *not* dissipation into a bath, and the earlier drafts' words "damped" and "decay rate" (with the bath-conventional symbol γ) invited exactly that misreading. A sub-threshold site has no stationary state at its partial energy (P4(i)); its excitation is virtual, and $\kappa_{\text{ret}} = \Delta E/\hbar$ is the rate at which the borrowed energy *reversibly returns to the common field* — the beat rate of a failed lock. The returned energy does not leave the system: it re-enters the exchange pool that the selection game redistributes, which is precisely why the shares can remain a martingale (energy that left irreversibly could not be a fair stake). Dictionary, to be maintained throughout: $\kappa_{\text{ret}} = \Delta E/\hbar$, the deficit-induced reversible return rate (state-dependent, vanishing at the shell); Δ, the drive–transition frequency mismatch (an independent parameter — two different mismatches, two symbols); Γ and γ, reserved for genuine dissipative rates (linewidths, bath damping, the $-2\gamma e$ term of §4). We state plainly what remains unproved: that the return rate is $\Delta E/\hbar$, and that reversible give-back may be modeled as a contraction eigenvalue in the site's reduced description, are *physically motivated ansätze* pending the microscopic derivation now listed in §9.4(viii).

The consequence is structural, not quantitative. The R. Adler entrainment average $\langle \sin\varphi \rangle \approx K/2\Delta\omega$ (Robert Adler's injection-locking equation [R. Adler 1946], not the trace-dynamics/collapse work of Stephen Adler cited above) — the standard expression for the systematic pull an oscillator exerts under injection, and the rich-get-richer channel here — is derived from *running* phase solutions of a limit-cycle oscillator, whose phase is a zero mode free to wind. Below threshold there is no limit cycle, no zero mode, and no running solutions: the entrainment tail is, *at linear order in this semiclassical analysis*, not small but absent. Whether a fully quantized common field admits a weak sub-threshold channel — via virtual-photon-mediated interaction — is not settled by the linear analysis; a controlled open-system calculation is required, and is listed among the open problems (§9.4). We therefore claim the tail absent to linear order and bounded by the layer width w beyond it, not absent unconditionally. A free clock — the prerequisite for biased energy extraction — emerges only where $\kappa_{\text{ret}} \to 0$, i.e. within a boundary layer

$$\Delta E \lesssim \hbar K \quad \Longleftrightarrow \quad w \equiv \frac{\Delta E_{\text{layer}}}{E_0} = \frac{K}{\omega}$$

of the shell, where K is the site–field coupling rate. Here P4(i) does decisive work: a discrete-level absorber has *no stationary state below the transition*, so partial winnings cannot be banked — sub-threshold excitation is virtual, its phase slaved to the common drive (identically for all sites, hence conferring no competitive advantage), its energy decaying on the uncertainty time $\hbar/\Delta E$. Level discreteness enforces threshold sharpness.

Three supporting results complete the argument. *(i) Feedback hierarchy:* from $\dot{e} = 2r\dot{r}$, a slaved site's uptake from the common field scales as $f\sqrt{e}$ — concave, hence share-*equalizing* (and equalizing toward the drive pattern, which by P1 is the Born pattern — harmless); its return κ_{ret}, e is exactly share-neutral; only autonomous (self-sustained) gain, confined to the layer, is share-amplifying. Rich-get-richer requires super-linear uptake, which a linear response cannot supply. *(ii) Quantified tolerance:* simulating the selection game with a rich-get-richer bias confined to a layer of width w gives Born deviations bounded at the percent level even for $w = 0.1$ and at the numerical floor for physical widths ($w = \Gamma/\omega \sim 10^{-6}$–$10^{-8}$ for atomic lines, $\sim 10^{-2}$ for broadband solids); by contrast, a bias with the (structurally absent) sub-threshold R. Adler tail would produce deviations an order of magnitude larger (Fig. 4).



atomic lines

broadband solids

(iii) Continuum absorbers (P4(ii)) were said in earlier drafts to evade the issue entirely, on the ground that their registration statistics are commit-rate-independent (Theorem 4), so that no biased late game exists to be played. That escape is withdrawn in v0.7 along with the fast-commit reading it rests on (§5.4). With commitment slow (§6.1), a continuum absorber runs to the absorbing boundary just as a discrete-level one does, and therefore traverses the same boundary layer and incurs the same $O(w)$ deviation. This is the more consistent position in any case: §6.1 already assigns broadband solids $w \sim 10^{-2}$ and percent-level deviations, which the evasion claim would have contradicted. The consequence is a gain rather than a loss — the predicted deviation applies to both absorber classes, widening the experimental target of §8 rather than exempting the most common detector type from it.

5.3 Theorem 3: the players cannot collude (noise locality)

Theorem 1 requires *independent* site noise. Vacuum fluctuations, however, are correlated over $\sim \lambda/2$; Theorem 3 computes exactly what such correlations do, and what protects real detectors.

Theorem 3 (Correlated-noise drift). *With $de_i = \sigma\sqrt{e_i}\, dW_i$ and $\langle dW_i\, dW_j\rangle = C_{ij}\, dt$,*

$$\mathbb{E}[\dot{s}_i] \;=\; \frac{\sigma^2}{E^2}\Big[\, s_i(\sqrt{e}\cdot C\cdot\sqrt{e}) \;-\; \sqrt{e_i}\,(C\sqrt{e})_i \Big],$$

which vanishes identically for $C = \mathbb{1}$ and is linear in C. (Proof: Appendix A.) For two sites with correlation c, the drift reduces to $\propto c\,\sqrt{e_1 e_2}\,(e_1 - e_2)$: rich-get-richer for $c > 0$, equalizing for $c < 0$.

Correlated noise is genuinely dangerous: with fully correlated noise ($C_{ij} = 1$) the brightest site wins with probability one, and in a resolved fringe geometry at spacing $d = \lambda/2$ a fully-radiative-noise model shows a +20% bright-favoured violation (a continuum of site separations defeats the apparent protection of the $\mathrm{sinc}(k\Delta r)$ zero at $\lambda/2$). What protects real detectors is not geometry but **locality of the dominant noise** (P3(b)): because the drift is linear in C, the deviation for a mixed kernel $C = (1-f)\mathbb{1} + fC_{\mathrm{rad}}$ scales linearly with the correlated fraction f (verified numerically over two decades of f; Appendix D), and $f \sim 10^{-6}$ for solid-state absorbers puts worst-case Born violations near 10^{-7}. Two further results feed the predictions of §8: the *vector* vacuum kernel is polarization-dependent — at $\Delta r = \lambda/2$ the correlation is $-3/2\pi^2 \approx -0.15$ for dipoles perpendicular to the separation but $+3/\pi^2 \approx +0.30$ parallel, so the deviation's *sign* rotates with polarization — and absorbers whose noise is *purely radiative* ($f \to 1$: free-space sub-wavelength atom arrays) are unprotected, making them the natural test bed.

5.4 Theorems 4 and 5: the finish line cannot distort (commitment)

Theorem 4 (Commit-rate independence). *Let registration fire at an arbitrary (possibly time-dependent) total rate, with conditional site-selection probability equal to the share s_i at the moment of firing — the property delivered by any rate law of the form* $\mathrm{rate}_i \propto e_i$. *Then, with s_i a martingale (Theorem 1) and the commit time T a stopping time,*

$$P_i = \mathbb{E}[s_i(T)] = s_i(0),$$

exactly, at any commit speed. Proof: optional stopping, applied at T rather than at absorption; the conditional pick probability being $s_i(T)$ makes P_i the expectation of a bounded martingale at a stopping time. ■

Theorem 4 reduces the registration question, *in the fast-commit regime*, to a single one: *is the commit rate linear in the site occupation?* The qualifier matters and is developed below. v0.7 held that commitment is slow compared with the game, so that the question did not arise; v0.8's corrected ladder (§6.1) finds the exposed window comparable to the commit time in silicon and longer than it in superconducting absorbers, so for continuum absorbers the question does arise, and the answer is not assumed (cf. P4) but derived:

Theorem 5 (Kinematic linearity). *In the n-quantum sector, under P2, a registration channel consuming $k \le n$ quanta at a site proceeds through exactly k absorption vertices, and its rate is therefore linear in the k-quantum coincidence weight $|\psi^{(k)}|^2$ at that site — never a higher power. In particular, for a single incident quantum,* $\mathrm{rate}_i = \Gamma\, e_i/E_0\,[1 + O(w^2)]$.

Argument. Each vertex is amplitude-linear (P2), so a k-vertex commit amplitude carries exactly one factor of the k-quantum amplitude. A hypothetical rate $\propto e_i^2$ in the one-quantum sector would require a two-absorption-vertex amplitude, which deposits $2\hbar\omega$ into a final state that does not exist: it is blocked by energy conservation *exactly*, not approximately. Absorption–emission combinations return quanta to the pool — they are the exchange dynamics of §4, not registration. The leading correction is a three-vertex (absorb–absorb–emit) path, suppressed by $(\Gamma/\omega)^2 = w^2 \lesssim 10^{-4}$ even for broadband silicon. Saturation corrections require occupied final states and vanish for a ground-state detector. ■

Simulations confirm the theorem pair and its converse (Appendix D): with rate $\propto s^\alpha$, the case $\alpha = 1$ reproduces Born at every commit speed tested, while $\alpha = 2$ ($\alpha = \frac{1}{2}$) produces bright-favoured (dim-favoured) deviations matching the closed-form fast-commit predictions ($P_1 = 0.941$ predicted, 0.938 observed, at $\alpha = 2$, two sites of Born weights 0.8/0.2) and relaxing toward Born as commitment slows.

What Theorems 4–5 are actually for (revised, v0.7; revised again, v0.8). Earlier drafts concluded here that the two registration classes of P4 deliver Born by *different* corollaries — discrete-level absorbers by first passage (Theorem 0 with absorption at $s = 1$), continuum absorbers by stopped rate-linear commitment (Theorems 4–5). v0.7 withdrew that division on the strength of §6.1's ladder, which then placed commitment at ns–μs, three to five orders above $\tau_{\text{game}} \sim 12$–$81\,\text{ps}$, so that the expected number of commitment opportunities during the game was far below one and first passage settled every event. v0.8 withdraws the withdrawal. The ns–μs figure is the avalanche-and-quench cycle; commitment as this paper defines it — first irreversibility — is available in a continuum absorber to any site whose share exceeds $\theta = E_{\text{gap}}/E_{\text{photon}}$, on the interband-vertex-plus-thermalisation clock of 10 fs–1 ps. The controlling quantity is still the expected number of commitment opportunities, but now during the leader's exposed climb from θ to 1, and §6.1 (v0.8) finds it of order unity for silicon and $\gg 1$ for superconducting absorbers.

Both classes therefore reach the Born weights through Theorem 4, not around it. Discrete-level absorbers have $\theta = 1$ and no exposed window, so first passage (Theorem 0) is the whole story there. Continuum absorbers have an exposed window, and Born survives in it exactly when the conditional pick at the moment of firing equals the share — the linear law of Theorem 5. Theorems 4–5 are load-bearing for class (ii), which is where v0.5 had them; v0.7's "robustness rider" framing is withdrawn.

What the linearity claim must cover is narrower than v0.7 feared, and this is the part of its argument that survives. The continuum-detector nonlinearities v0.7 cited — SNSPD detection efficiency falling *exponentially* at low photon energy under the vortex-crossing model, *supralinear* response in both SPADs and SNSPDs — are properties of the downstream cascade: avalanche triggering, vortex crossing, hotspot growth. That cascade is photon-agnostic: a thermally generated carrier traverses it identically, so it cannot carry which-site information and cannot be the commit law that enters Theorem 4. The law that enters is the rate at which an exposed site's virtual holding converts to a real excitation — the interband absorption vertex — and Theorem 5's own argument, each vertex amplitude-linear under P2, is an argument about exactly that vertex. The downstream nonlinearities then act as a site-independent thinning, absorbed into efficiency. The simulation separating commit *speed* from commit *law* remains the right diagnostic: with a deliberately wrong (Arrhenius) commit law on the ten-site configuration, the bright site wins with probability 0.719 against Born's 0.500 when commitment is fast, and 0.504 when it is slowed to the separation v0.7 assumed. Under v0.8 the paper operates *between* those two points, not at the second, so a wrong law would not be harmless where the paper operates; what protects the statistics is that the law at the vertex is linear, not that commitment is slow. A discrete-exchange simulation with the commit channel gated at θ is owed (§9.4(xi)).

P4's two-class taxonomy thus survives as a statement about *how the full-quantum threshold is enforced* — spectroscopically in class (i), by clause (a) as a premise in class (ii) — and as a statement about *whether an exposed window exists*: none for $\theta = 1$, one of width $1 - \theta$ on the share axis otherwise.

v0.9: the gate, tested, and the reading that removes it. The preceding paragraphs assume reading A of §9.4(xi). A discrete-exchange simulation with the commit channel gated at θ (`g3_drain_tests/threshold_gated_commit.py`; hazard $\lambda f(e_i)$ per step for exposed sites only, the first to fire taking the whole quantum, first passage otherwise) tests both halves of the v0.8 argument. The ungated rows ($\theta = 0$) reproduce Born at every commit speed for the linear law, to Monte-Carlo precision: Theorem 4 as stated. Every gated row does not. With the expected number of commits during the winner's exposed leg $r = \lambda\, t_{\text{exp}}$, the linear-but-gated law biases an 80/20 split by +0.04 at $r = 1$, +0.07 at $r = 3$ and +0.12 at $r = 10$, and the ten-site configuration (bright site Born 0.500) by +0.04, +0.09 and +0.19 — because Theorem 4 needs the conditional pick to equal the share for *every* site, and a gate sets it to zero for the unexposed ones. So linearity does not rescue the exposed window; only $r \ll 1$ does, which corrects the v0.8 sentence above. And the exposure itself is far longer than the v0.8 estimate: in this engine the eventual winner spends 40–90% of the game above θ (two-site: 97 of 107 steps at $\theta \le 0.72$, 52 of 108 at $\theta = 0.94$; ten-site: 83% at $\theta = 0.45$, 37% at $\theta = 0.94$), because the stakes-scaled exchange slows as the losers drain, so $t_{\text{exp}} \approx \tau_{\text{game}}$ and reading A in silicon sits at $r \sim 10$–10^4, where the bias saturates near +0.1 to +0.2. Under reading B there is no gate, the $\theta = 0$ rows are the physical case, and Theorem 4 with the linear vertex law of Theorem 5 is the whole of the registration argument. That is the reading v0.9 adopts (header). The consequence for what the game is *for*: Theorems 1–3 describe the reversible sub-threshold dynamics and its fairness — what the substrate must not do to the shares before commitment — and Theorem 4 says that, given a linear hazard, *when* commitment happens is irrelevant. The Born frequencies enter through the hazard's linearity in the share, i.e. through P1 with Theorem 5, as §1 and §9 already state.

5.5 Synthesis

The four theorems close the four channels by which a physical selection process could deviate from a fair game — biased dice (Th. 1), a self-amplifying leader (Th. 2), colluding players (Th. 3), and a distorting finish line (Th. 4–5) — and each closure is purchased by an identified physical premise rather than by fiat: P2 buys Theorems 1 and 5, P1+P4 buy Theorem 2, P3 buys Theorem 3. The fairness that dynamical-reduction models must engineer into their noise is here a consequence of detector physics. By the engine theorem, the outcome probabilities are the initial energy shares; by P1, the initial energy shares are $|A_i|^2/\sum_j |A_j|^2$. That is the Born rule — with every step either a stated premise or a proved theorem, and with each premise's failure mode a computable, testable deviation (§8).

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6. Reproducing quantum statistics

The theorems of §5 are exact in their stated limits; a physical detector must realize those limits within its actual hierarchy of rates, and the mechanism must extend beyond single quanta to the full counting statistics of quantum optics. This section verifies both.

6.1 The timescale window in a real detector

We take a silicon single-photon avalanche diode at $\lambda = 500\,\mathrm{nm}$ as the working example ($\omega = 3.8 \times 10^{15}$ rad/s; $E_0 = 2.5\,\mathrm{eV}$). The relevant ladder:

- *Exchange step:* by fluctuation–dissipation, the noise driving the game has variance rate set by the final-state width, $\Gamma \sim 1.5 \times 10^{13}$–$10^{14}\,\mathrm{s}^{-1}$ (widths of 10–100 meV): one exchange step per 10–70 fs.
- *Game duration:* the selection must transport log-shares a distance $\ln N$ diffusively, giving $\tau_{\mathrm{game}} \sim 2\ln^2 N/\Gamma \approx$ 12–81 ps for $N \sim 5 \times 10^{10}$ sites in the diffraction volume, across the stated Γ range (a device-specific estimate, not a substitute for a detector-specific open-system calculation). (Only $\ln^2 N$ enters; the answer is insensitive to the coarse-graining of "sites.")
- *Losses, and the protected range:* silicon's indirect gap puts radiative loss at microsecond scales, and multiphonon dissipation of a *virtual* holding is blocked by the absence of sub-gap final states — but only while the holding lies below the gap. A site's share s_i carries $e_i = s_i E_0$, and no real final state exists for $s_i < \theta = E_{\mathrm{gap}}/E_0 = 0.45$ at 500 nm. Below θ the same discreteness that sharpens the threshold (Th. 2) protects the stakes, and the loss fraction during the game is $\lesssim 10^{-5}$. Above θ a real electron–hole pair is available, with $e_i - E_{\mathrm{gap}}$ going to phonons, and the interband vertex plus thermalisation — 10 fs–1 ps in silicon (literature-typical: photon-annihilation irreversibility fs–100 fs; which-site distinguishability 10 fs–1 ps; field-driven seed unrecoverability \$ \$1 ps) — is the first irreversibility. That, not the avalanche, is τ_{commit} in the sense of Theorem 2 and §4. The ns–μs quoted in versions before v0.8 is the avalanche-and-quench cycle: record-writing, three to six orders downstream. Note also that $E_0 = 2.2\,E_{\mathrm{gap}}$: the closed pot could pay for two real sub-gap excitations, so at this wavelength the gap does not enforce exclusivity; P4(a) does, as a premise (§2). The boundary $E_0 = 2E_{\mathrm{gap}}$ sits at 553 nm.
- *Exposed window:* only a site above θ can commit. For $\theta \geq 1/2$ that is at most one site under a closed pot, during its final climb from θ to 1; at 500 nm, with $\theta = 0.45$, two sites can be exposed at once, which is the exclusivity question P4(a) settles by premise. On the same diffusive scaling as τ_{game} the leader's exposed leg takes $t_{\mathrm{exp}} \sim 2\ln^2(1/\theta)/\Gamma \approx 13$–$84\,\mathrm{fs}$ — about one exchange step, where the continuum scaling is least reliable (a discrete exchange step moves a share by at most a quarter of the smaller holding, so several steps are needed, which lengthens t_{exp}).

- *Pattern-degrading transport:* below threshold, coherent transport of the excitation *is* the slaved polariton wave — it carries the Born pattern rather than diffusing it. Incoherent diffusion requires real population, which exists only after commitment: spread $\sqrt{2D\tau_{\text{commit}}} \sim 8\,\text{nm}$ against a micron-scale pattern.

The two lower rungs close with margin: $1.7\,\text{fs} \ll 10\text{–}70\,\text{fs} \ll 12\text{–}81\,\text{ps}$, and the middle separation $\tau_{\text{game}}/\tau_c = 2\ln^2 N \approx 1200$ is *independent of* Γ — both rungs scale as $1/\Gamma$ — so it survives cooling, a change of material, or any other change in the electronic relaxation rate. (A robustness result first stated in v0.8: the thermal scattering that supplies the white noise can be frozen out without the noise ceasing to be white on the game's own clock.) The top rung does not close with margin. Versions before v0.8 wrote $12\text{–}81\,\text{ps} \ll \text{ns–}\mu\text{s}$ and claimed two to four orders of magnitude to spare at every rung; with commitment placed where this paper's own definition puts it, the comparison is $t_{\text{exp}}/\tau_{\text{commit}} \approx 0.01\text{–}8$: **marginal**, a range straddling unity by about an order of magnitude on each side, closing toward safe only near the band edge ($\theta = 0.90$ at 1000 nm gives 2×10^{-4}–0.14). For a superconducting absorber the rung inverts: NbN at 1550 nm has $E_{\text{photon}}/E_{\text{gap}} \approx 500$, $\theta \approx 0.002$, an exposed leg of 6.3 log-units ($\approx$ 79 exchange steps, 0.8–1.6 ns at $\tau_{ep} \sim 10\text{–}20\,\text{ps}$) against a hotspot commit of 10–50 ps — a ratio of 16–160 across the whole input range. The consequence for §5.4 is that Theorems 4–5 are load-bearing for continuum absorbers, and what protects the statistics in the exposed window is the linearity of the vertex law, not the slowness of commitment. What the window would cost if that law were not linear is computable: instantaneous commitment above θ would send a Born 0.80/0.20 split to $P_1 = 0.87$ at 1000 nm and to 1.00 at 650–800 nm — this section's own $O(1-\theta)$ stopping bias. No published asymmetric-split Born-ratio test below 553 nm was located (2026-09 search), so the percent-level agreement one would expect from calibration practice is an inference, not a measurement; the coincidence branch, however, is bounded at $g^{(2)}(0) = 2 \times 10^{-3}$ by heralded 505 nm data taken with silicon SPADs, where both arms of a balanced split sit above θ from the outset. Both orderings of the photon coherence time against the game are benign. A long wavepacket continually re-pins the shares to the instantaneous Born pattern — a re-preparation, not a distortion — and predicts a click-time density proportional to $|\text{envelope}(t)|^2$, i.e. time-resolved Born statistics, as observed. A short pulse leaves the game to run drive-free after capture, the clean martingale regime. (We note without weight that τ_{game} falls in the tens-of-picoseconds range of measured SPAD timing jitter.)

v0.9 — the top rung under reading B, and what the discrete engine says about reading A. Under reading B (adopted; header), τ_{commit} is the golden-rule absorption time at a site, its hazard linear in the site's holding and available from every site, and the rung $t_{\text{exp}} \ll \tau_{\text{commit}}$ is not a condition: Theorem 4 holds at any commit speed for an ungated linear hazard (§5.4, v0.9). Under reading A, the v0.8 estimate $t_{\text{exp}} \approx 13\text{–}84\,\text{fs}$ — one exchange step — is replaced by a measurement in this paper's own discrete engine: the eventual winner spends 40–90% of the game above θ, because the stakes-scaled exchange slows as the losers drain. So $t_{\text{exp}} \approx \tau_{\text{game}}$, the whole-game accounting $\tau_{\text{game}}/\tau_{\text{commit}} = 12\text{–}8100$ is the right one for reading A in silicon — inverted, not marginal — unless the continuum kernel (the discretisation question of §9.4(iv)) is the physical one, and with the gate fully open a gated commit law biases asymmetric splits by of order +0.1 (§5.4). Reading A in silicon therefore predicts deviations that calibration practice would be expected to have noticed, though no published asymmetric-split test below 553 nm was located; its no-P4(a) version is already bounded by heralded 505 nm $g^{(2)}(0) = 0.0023$ to $t_{\text{exp}}/\tau_{\text{commit}} \lesssim 10^{-4}$. Those are the grounds, with the microscopic model's own construction, on which v0.9 adopts B.

Why the boundary sits at unity. The game terminates when one site holds the whole quantum ($s_i \to 1$); the simulations (Appendix D) reproduce Born to the Monte-Carlo floor at this absorbing boundary, whereas stopping at any finite share introduces a computable $O(1 - \text{threshold})$ bias. The unit boundary is not a numerical idealization but the temporal face of Theorem 2: below the shell a site has no autonomous phase and cannot sustain a lock, so a partial quantum cannot be banked, and the irreversible lock engages only as a site assembles essentially the full quantum. The residual gap $1 - \theta_{\text{lock}} = O(w)$, with $w = K/\omega$ the boundary-layer width, is then the physical origin of the predicted deviations — negligible for atomic lines ($w \sim 10^{-6}$), percent-level for broadband solids ($w \sim 10^{-2}$). The oversized threshold of a crude stopping rule and the physical deviation are the same expression evaluated at different gaps.

The temporal structure of an event. The ordering of the ladder is not incidental. Capture is bounded below only by the energy–time uncertainty, $\tau_{\text{capture}} \gtrsim \hbar/\Delta E$; for a massless incident quantum, which carries no rest-mass Compton clock, this bound is saturated at $\tau \sim 1/\omega$ — the fastest an interaction can proceed — so the squared-amplitude pattern is imprinted essentially instantaneously and faithfully, before any slower dynamics can bias it. The lock then engages *reversibly* on the fast synchronization scale; irreversibility enters when an exposed site's holding becomes a real excitation — in a semiconductor, at the interband vertex and thermalisation, 10 fs–1 ps — and the record is written afterwards by the reservoir-powered amplification, which cannot un-write a thermalised carrier. (Versions before v0.8 placed irreversibility at the record-writing step; v0.8 moves it up the ladder to where the paper's own definition of commitment puts it.) Two consequences follow. First, by the commit-rate independence of Theorem 4 — given the linear vertex law of Theorem 5 — the rate of the commit step drops out of the outcome weights: dissipation sets *when* an outcome finalizes, never *toward whom*. Second, the reduced site-energy fails to be conserved (§5.1) precisely because energy leaves the site subsystem into the bath at this step — which locates the conserving field+detector+bath derivation (§9.4) as one in which the fast capture–selection dynamics is nearly closed and the slow registration is where energy irreversibly flows out. We offer this temporal picture as interpretation consistent with Theorems 2 and 4, not as an independent theorem; the controlled open-system derivation is an open problem (§9.4).

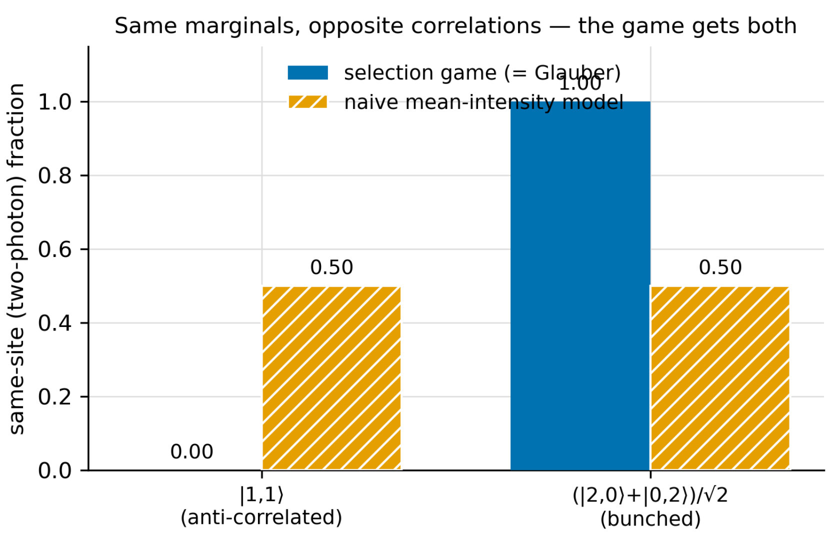
6.2 Multi-quantum states: Glauber counting

For n incident quanta the medium carries the *joint* amplitude (P5) — for two, the symmetric pair registry $\psi(i,j)$ — and detector stakes are drawn from it sector by sector. A one-photon absorber plays for the single-quantum marginal $m_i = \sum_j |\psi(i,j)|^2$. Upon the first commitment, at site i^*, the registry renormalizes (the same conditional update that §7 uses across entangled wings), and the second game runs on the conditional row $|\psi(i^*,j)|^2/m_{i^*}$. Each game is Born-exact by §5, so the joint distribution is marginal times conditional:

$$P(i,j) \;=\; m_i \cdot \frac{|\psi(i,j)|^2}{m_i} \;=\; |\psi(i,j)|^2$$

— Glauber's joint photodetection distribution, from the same composition property that will produce the Bell correlations.

The discriminating test is a pair of states with *identical mean intensities but opposite correlations*, for which any intensity-only model returns identical predictions. For two site-localized modes: the anti-correlated state $|1,1\rangle$ (one quantum at each site) and the bunched state $(|2,0\rangle + |0,2\rangle)/\sqrt{2}$ both give 50/50 marginals. The simulated game yields perfect anti-bunching for the first (the two quanta *never* register at the same site; split fraction 1.000) and perfect bunching for the second (same-site fraction 1.000), reproducing the Glauber predictions exactly, where the intensity-only model predicts an uncorrelated 50/50 split for both (Fig. 5).



For a generic (randomly generated) three-site pair state, the simulated joint distribution matches $|\psi(i,j)|^2$ entrywise to 2×10^{-3}, the statistical floor of the run.

6.3 Nonlinear-intensity detectors and ρ-linearity

A two-photon absorber — a site with no level at $\hbar\omega$ but one at $2\hbar\omega$ — commits through exactly two absorption vertices (Theorem 5), so its amplitude carries exactly one factor of the *pair* amplitude at the site, and its rate is

$$\text{rate}_i \;\propto\; |\psi(i,i)|^2 \;=\; G^{(2)}(i,i),$$

the standard two-photon POVM: nonlinear in intensity, linear in the pair-sector weight. The registry structure is not optional decoration; it is forced by consistency. On the state $|1,1\rangle$ the pair diagonal vanishes, $\psi(i,i) = 0$, and the game correctly predicts *no* two-photon registrations at either site — where a model that squared the mean intensity would falsely predict clicks at both. The simulation confirms both this null and the 50/50 two-photon statistics of the bunched state.

The general statement assembles from Theorem 5 and the convex structure of the ensemble. Every allowed commit channel's rate is linear in its coincidence weight $G^{(k)}$; every $G^{(k)}$ is a linear functional of ρ; per-configuration outcome statistics are Born (§5); and a mixed state, realized as an ensemble of substrate configurations, averages its predictions affinely (verified numerically to 6×10^{-4}). Hence *all* outcome statistics are affine in ρ — for arbitrary detectors, in every sector — with two immediate consequences: predictions depend on a mixed state only through ρ (decomposition-independence), and the ground is laid for the no-signaling results of §7. The framework thus reproduces the POVM structure of photodetection theory, with the selection mechanism running beneath it rather than instead of it.

7. Entangled pairs and no-signaling

7.1 The joint game

The decisive test of any outcome-selection mechanism is the entangled pair, where Bell's theorem forbids the correlations from arising in independent local games. Consider $|\Phi^+\rangle = (|HH\rangle + |VV\rangle)/\sqrt{2}$ distributed to stations A and B with analyzers at angles a, b, each followed by a two-port detector. Two structural facts dictate the mechanism's shape. First, each wing's *local* wave is unpolarized — its reduced state gives 50/50 shares in every basis — so by §5 the local games can produce only the correct *marginals*, which carry no angle information. This is as it must be: local statistics in a Bell experiment are coin flips. Second, the joint weights $P(\pm\pm) = \frac{1}{2}\cos^2(a-b)$ reside in the *joint* amplitude, which by P5 is physically instantiated: the pair was created as a single excitation of the shared medium, and the two waves retain a stored phase relationship — the same registry structure that carried the pair amplitude $\psi(i,j)$ in §6.2. This shared structure is explicit and real — a correlation that local bookkeeping ignores, not a hidden variable.

The selection process is then two-stage. Both local games run concurrently. Whichever wing *commits first* — the ordering supplied by the foliation of P6 — undergoes a lock that renormalizes the pair registry: the joint amplitude re-forms around the realized outcome, as dynamics rather than decree, and the far wing's shares are thereby reset to the conditional Born weights (e.g. $P(B{=}+\,|\,A{=}+) = \cos^2(a-b)$). The far wing's game then completes on the updated shares. Because a conditional reset is simply a new initial condition, and the theorems of §5 hold for *any* initial condition, both factors are exact:

$$P(i,j) \;=\; \underbrace{m(i)}_{\text{marginal}} \times \underbrace{P(j\,|\,i)}_{\text{conditional}} \;=\; |c_{ij}|^2,$$

independently of which wing commits first — and order-independence of the statistics is precisely the no-signaling property. Simulation of the full stochastic process (two concurrent games, stochastic commit order, conditional resync; Appendix D) gives CHSH $S = 2.820$ at the standard angles against the quantum value $2\sqrt{2} = 2.828$; $E(a_1, b_1) = 0.706$ vs 0.707; the A marginal flat under change of B's setting (±0.007, consistent with zero); and identical correlations conditioned on commit order (0.717 vs 0.706). We emphasize that reproducing $2\sqrt{2}$ is a *consistency check, not a result* — any Born-faithful mechanism inherits it. The content of this section lies in what follows.

7.2 The fidelity law

Degrading the conditional update — far-wing shares reset to $\eta \cdot \text{(conditional)} + (1-\eta) \cdot \text{(current)}$ — yields $E(a,b) = \eta \cos 2(a-b)$ and hence

$$S = 2\sqrt{2}\,\eta,$$

verified across the sweep ($\eta = 1,\ 0.9,\ 1/\sqrt{2},\ 0.5 \to S = 2.820,\ 2.533,\ 2.000,\ 1.421$): the local-realism bound is crossed exactly at $\eta = 1/\sqrt{2}$. The *resync fidelity* η is thereby promoted from interpretive gloss to physical parameter, and existing experiments measure it: photonic CHSH values near 2.8 over 10^2-km separations imply $\eta \gtrsim 0.99$. Whatever the substrate's update mechanism is, it is near-perfect at terrestrial scales. (§8.7 proposes a *controlled* measurement of η: deliberately injected broadband cross-station noise leaks into the correlators only through registry infidelity, with a computable, angle-independent signature.)

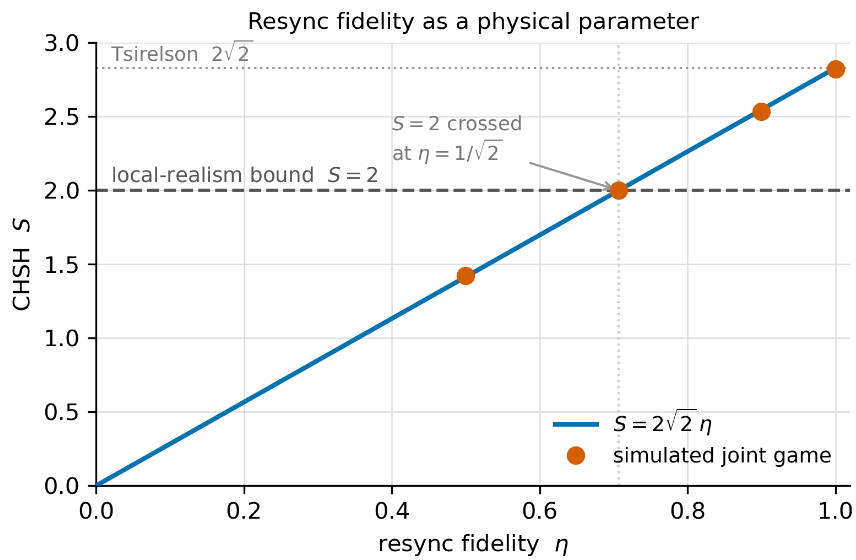


Figure 3: **Figure 6.** The resync-fidelity law $S = 2\sqrt{2}\,\eta$. Degrading the conditional update to fidelity η scales the CHSH value linearly; the simulated joint game (points) tracks the law, crosses the local-realism bound $S = 2$ exactly at $\eta = 1/\sqrt{2}$, and reaches the Tsirelson value $2\sqrt{2}$ at $\eta = 1$. The fidelity η is thereby a measurable physical parameter, not an interpretive gloss.

The single-quantum sector fixes η rather than measuring it (v0.7). With P5 broadened (§2) to cover any state whose support spans causally disconnected candidate sites, the beam-splitter is the fidelity law's degenerate case, and it behaves qualitatively differently. Registration consumes the full quantum (P4(a)) out of a closed pot, so a surviving fraction $(1-\eta)$ in the far arm would leave the ledger holding $\hbar\omega\,[1+(1-\eta)/2]$: not a degraded update but a non-conserving one. The mechanism therefore predicts **exact** anticorrelation for a true one-quantum input, with no adjustable fidelity — the η of Figure 6 is a parameter of the entangled sector alone.

This changes what anticorrelation data can and cannot do for the theory, in both directions. It cannot be used to *measure* η: a nonzero $g^{(2)}(0)$ cannot be read as registry infidelity, because the mechanism forbids that channel outright. What it does instead is expose the mechanism to a clean falsifier. Any coincidence rate above what the source's own multi-photon content accounts for is a direct contradiction, not a fitted degradation — so the constraint tightens with every improvement in heralded single-photon sources, and it does so without a free parameter to absorb the result. The historical measurement [Grangier et al. 1986] and its modern successors are, on this reading, tests of P4(a) and P5 jointly, and the relevant systematic is entirely the source characterization rather than anything in the detector model.

The asymmetry also strengthens §7.3's argument that $\eta = 1$ exactly. That section gives a geometric reason — a shared frame rotates, and rotations do not attenuate with distance. Conservation now supplies a second, independent reason in the single-quantum sector. If the substrate's update mechanism does not distinguish a delocalized one-quantum amplitude from a two-quantum registry, then η is pinned in the sector where conservation bites, and the entangled sector inherits the value rather than being free to differ. Whether the update mechanism is in fact sector-blind is not established here, and is added to the open problems of §9.4.

7.3 The resync mechanism and the speed bounds

What kind of process could realize the faithful update that P5 asserts? The candidates divide into *propagating* influences (finite $v \gg c$ in the preferred frame) and *global* re-equilibration. The picture adopted here is the latter, in geometric form: the pair is one substrate object whose correlation is a single shared internal orientation — a phase frame anchored at the source event — and the first lock *rotates that shared frame*. The changed degree of freedom is an angle, not a field configuration traversing the intervening space; the apparent superluminality is a category error of the same family as phase-velocity and pattern motions (the intersection point of closing scissors), which move arbitrarily fast precisely because nothing physical transits. Three consequences discriminate this picture: (i) experiments bounding the "speed of quantum influence" can only ever return growing lower bounds — currently $> 5.4 \times 10^4\, c$ for a CMB-like preferred frame (Salart et al.) and $\geq 1.38 \times 10^4\, c$ (Yin et al.) — and the rotation picture predicts *no floor will ever be found*, whereas any propagating variant must eventually exhibit one; (ii) fidelity does not degrade with distance — rotations do not attenuate — consistent with undiminished violation at satellite scales, and making $\eta = 1$ natural rather than tuned; (iii) "before-before" configurations with moving detectors (Zbinden et al.; Stefanov et al.), in which each wing is first in its own rest frame, are dissolved: the ordering that matters is carried by the foliation at the shared object, not by either detector's frame — consistent with the null results of those experiments.

7.4 No-signaling as a theorem, and the rigidity of the statistics

The traditional logic (Gisin) runs: no-signaling is imposed, and it forces the linear/martingale structure of any acceptable stochastic dynamics. Here the arrow reverses. Theorem 1 (fair shares) and Theorem 5 (kinematically linear commitment) are *derived* from the premises, they entail statistics affine in ρ (§6.3), and affine local statistics entail that no operation at one wing moves the marginals at the other: **no-signaling follows as a theorem** — but one *conditional on the shared-registry premise P5* (with faithful update $\eta = 1$), which supplies the affine structure in the first place. The logical order is nonlocal ontology (P5/P6) $\to$ Born-weighted local games $\to$ affine statistics $\to$ no-signaling; this is a *consistency* result, not a derivation of no-signaling from local detector physics. It holds exactly at the statistical level over an explicitly nonlocal ontic structure.

The inverse direction quantifies how rigid this is. Introduce a hypothetical commit nonlinearity at wing A, $f(s) = s^{1+\varepsilon}$, and prepare the partially entangled state $\sqrt{p}\,|00\rangle + \sqrt{1-p}\,|11\rangle$. The fast-commit limit gives the far-wing marginal in closed form,

$$M(B{=}0;\alpha) = p\,\frac{\cos^2\!\alpha\; s_+^{\varepsilon} + \sin^2\!\alpha\; s_-^{\varepsilon}}{s_+^{1+\varepsilon} + s_-^{1+\varepsilon}}, \qquad s_+ = p\cos^2\!\alpha + (1{-}p)\sin^2\!\alpha,$$

which acquires a dependence on A's setting α — a signal — of magnitude $\Delta M \approx K(p)\,\varepsilon\,P_{\text{first}}$ with $K(0.8) \approx 0.18$ and $P_{\text{first}} = \frac{1}{2}$ for symmetric stations (full-game simulation: $\Delta M = +0.044$ at $\varepsilon = 0.5$ against the ceiling $+0.089$; -0.0002 at $\varepsilon = 0$). Two exact zeros structure the phenomenology: $\varepsilon = 0$ (any state — the theorem), and $p = \frac{1}{2}$ (any ε — a symmetry protection at maximal entanglement). The latter has an experimental moral: *maximally entangled Bell tests, which is to say nearly all Bell tests, are the least sensitive configuration for this physics*; the sensitive test is partial entanglement with asymmetric bases, a pointer developed in §8.

7.5 What is derived and what is postulated

Derived: that concurrent local fair games with rate-linear commitment convert the joint amplitudes into exactly Born-weighted, order-independent, no-signaling joint outcomes — the composition property doing for two wings what it did for two quanta in §6.2. Postulated: P5 itself — that the medium carries the joint amplitude between the wings and renormalizes it faithfully ($\eta = 1$) — and the ordering foliation P6. This section is therefore a *consistency proof* of the selection mechanism with the entanglement ontology of P5–P6, not a derivation of entanglement; the nonlocality is not explained away but located — in the shared registry and the preferred foliation — and η, currently an input measured to hold at the percent level, is the natural next target for derivation, with §7.3's rotation picture indicating why its value should be exactly one.

8. Predictions and falsifiability

8.1 The deviation ledger

A mechanism, unlike an axiom, can fail — and the premises of §2 delimit exactly where. Table 1 records every candidate deviation channel this program has generated, including two that the theory itself subsequently eliminated. We display the retractions deliberately: a mechanism should generate candidate failure modes and then be held to its own theorems, and the pattern of closures is itself evidence about the surviving channels.

Table 1 — Born-deviation channels and status.

Channel	Premise at risk	Status
Soft locking threshold (broadband absorbers, layer width $w = \Gamma/\omega \sim 10^{-2}$)	P4	**Retracted.** Artifact of imposing a hard threshold on a continuum absorber; Theorem 4 makes the statistics commit-rate-independent.
Nonlinear commit rate (cooperative/two-carrier registration), any sector	P4	**Retracted.** Kinematically blocked by Theorem 5; residual $\varepsilon \lesssim w^2 \lesssim 10^{-4}$.
Correlated vacuum noise (radiative-dominated absorbers, e.g. sub-λ atom arrays)	P3(b)	**Live.** Deviations $\propto$ correlated noise fraction f; large only as $f \to 1$.
Pre-established absorber coherence (phase-locked ensembles)	P3(a)	**Live** (with the caveat that maintaining coherence requires driving, which exits the single-quantum sector; §8.6).
Threshold-gated commitment (continuum absorbers, $E_{\text{photon}} > E_{\text{gap}}$): commitment open to sites with $s_i > \theta = E_{\text{gap}}/E_{\text{photon}}$ during the leader's exposed climb — §6.1 (v0.8)	P4(a), P4(b)(ii); Theorem 5's linearity	**Reading A only (v0.9).** Under the adopted reading B no gate exists and Theorem 4 closes the channel. Under reading A it is open for 40–90% of the game in this paper's discrete engine and biases the outcome even for a linear law (a gate is not Theorem 4's law): +0.04 at one expected commit per exposed leg, +0.12 to +0.19 at ten, on 80/20 and ten-site splits (§5.4). The no-P4(a) version is bounded by heralded 505 nm $g^{(2)}(0) = 0.0023$ to $r \lesssim 10^{-4}$. Retained as the discriminator between the readings (§8.6(vii)), not as a prediction of the paper.

Two of the surviving channels are failures of P3 — the Stage-2 fairness conditions — and neither afflicts any detector in experimental use. The third, added in v0.8, sits at the P4 boundary and *does* concern detectors in use; it is the price of placing commitment where the paper's own definition puts it, and it is bounded rather than excluded by the data the paper is aware of. In v0.9 it is reclassified: closed under the adopted reading B, open and over-predicting under reading A, hence the discriminator between the two rather than a prediction of the paper.

8.2 The port decomposition: why the entire experimental record is protected

Deviations of the surviving type bias *which site within a detector patch* registers. But the outcome that experiments record — and that carries correlation information — is the *port* (which analyzer output fired). Summing Theorem 3's drift over the sites of each port (cross-port noise correlations vanish for macroscopically separated ports) gives, for the port share S_A,

$$\dot{S}_A \ \propto\ q_B\, S_A - q_A\, S_B, \qquad q_P = \sum_{i,j \in P} C_{ij} \sqrt{e_i e_j}\,,$$

with the immediate consequence: **if the two ports have matched collective structure** ($q_A/S_A = q_B/S_B$ — same geometry, same internal correlation) **the port share is drift-free, and port statistics are exactly Born even when within-port statistics deviate arbitrarily.** Within-port deviations are invisible to every recorded observable and can never signal. All experiments on record use structurally symmetric ports; combined with $f \sim 10^{-6}$ for their detectors, the existing record is protected with roughly six orders of magnitude to spare. The mechanism is not threatened by any measurement yet performed — and, less comfortably, it has not yet been *tested* by one either. The remainder of this section designs the test.

8.3 The live configuration

Let the two ports differ in collective structure — one a dense sub-wavelength array, the other sparse; or one phase-locked to an external reference. Simulation of the port game (Appendix D) gives a bias linear in the mismatch of the within-port correlated noise fraction, $\kappa \approx -\Delta c_{\text{eff}}$, with port statistics

$$P'(S) \ \approx\ S + \kappa\, S(1-S)$$

(the internally correlated port *suppressed* — a Jensen-inequality effect: correlated noise inflates a port's energy variance, and shares are concave in energy), together with a state-dependent no-click rate in the strong-correlation limit.

An essential caution: for radiative-dominated arrays, standard quantum mechanics *itself* predicts collective modifications of absorption — the response is governed by dressed (sub- and superradiant) modes, not independent atoms [Asenjo-Garcia et al. 2017; Javanainen et al. 2014; Javanainen & Ruostekoski 2016] — and the collective response is not hypothetical but measured: a single sub-wavelength monolayer of a few hundred atoms shows subradiant line narrowing and mirror-grade reflectivity [Rui et al. 2020], controllable at the single-photon level [Srakaew et al. 2023]. Deviation from *naive per-site* weighting is therefore ordinary cooperative optics and tests nothing. The framework's claim is a deviation from the *dressed-mode Born prediction*; the quantitative confrontation of the port game with the collective-mode calculation is open work, and the discriminator below is designed so that this subtlety cannot contaminate it.

8.4 A tabletop discriminator

Send single photons at a variable splitting ratio $S : (1-S)$ into the two ports of a detector with deliberately mismatched collective structure, the ports spatially separated, and record the conditional port statistics $P'(S)$. For *any* local POVM — i.e., for standard quantum mechanics with arbitrary collective physics inside each port — the cross-port matrix elements vanish and $P'(S)$ is exactly affine in S: a straight line whose slope encodes the (arbitrary, possibly collective) port efficiencies. The framework predicts *curvature*, $\kappa S(1-S)$, plus a state-dependent no-click anomaly. The signature is thus insensitive to the dressed-mode physics inside each port: any nonzero curvature (beyond calibrated nonlinearities) discriminates against the linear-POVM structure of quantum mechanics itself. No entanglement, no timing, no Bell configuration — a single-photon

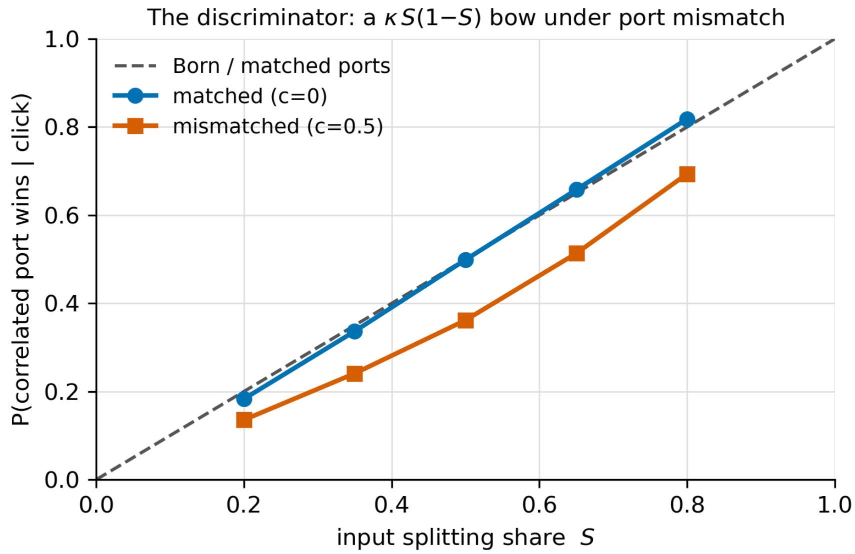


Figure 4: **Figure 7.** The tabletop discriminator. Conditional probability that the correlated port registers, versus the input splitting share S. Matched ports ($c = 0$) track the Born diagonal; deliberately mismatched within-port noise correlation ($c = 0.5$) produces the predicted $\kappa\, S(1 - S)$ bow, suppressing the internally-correlated port. Any nonzero curvature beyond calibrated efficiency effects discriminates against the linear-POVM structure of standard quantum mechanics.

source, a variable splitter, and counting statistics probe the mechanism's most radical corner at the $|\kappa|$ level, with candidate implementations (tweezer-array ports at sub-λ spacing, $f \to 1$) reaching $|\kappa| = O(10^{-1})$ if the live channel is real. The ingredients are separately demonstrated: sub-wavelength atomic arrays with measured cooperative optical response, down to the single-photon level [Rui et al. 2020; Srakaew et al. 2023], and single-shot registration of one absorbed photon by a cold-atom ensemble, at efficiencies from $\sim 20\%$ in free space to $\gtrsim 90\%$ with cavity assistance [Paris-Mandoki et al. 2017; Yang et al. 2022; Vaneecloo et al. 2022]. What remains unbuilt is the combination: an array operated as a *calibrated absorption-counting port*. One distinction matters in assembling it: Rydberg blockade realizes the $f \to 1$ limit taken all the way to a single effective absorber — erasing the within-port site competition that the port game concerns — so blockaded superatoms supply the readout existence proof, while the deviation medium proper is the un-blockaded array of competing sites.

8.5 The fork

If the live channel is real, its consequences do not stop at Born deviations. A mismatched-port detector at one wing of an entangled pair shifts the far wing's marginals — and because the asymmetry resides in the *detector*, the maximal-entanglement protection of §7.4 does not apply: $M(\alpha) = \frac{1}{2} + \delta \cos 2\alpha \cdot P_{\text{first}}$, a setting-dependence, i.e. in-principle superluminal signaling. Two readings are open, and we state them as a fork rather than resolve it prematurely. On the *radical reading*, the prediction is genuine; nothing in existing data excludes it, because the configuration has never been built. On the *protective reading*, a deeper fairness principle — for instance, a compensating renormalization of uptake and decay by the same substrate dynamics that correlates the noise — closes this channel as kinematic linearity closed the commitment channel, and the framework becomes *derivably* equivalent to quantum mechanics for all detector configurations, which would itself be the strongest form of the program. Our judgment favors the protective reading: twice already in this work, channels left open by first analysis were closed by theorems (Theorems 4 and 5, retiring the first two ledger entries), and the pattern favors closure. But the framework as here formulated cannot decide, and the experiment of §8.4 can: measured curvature confirms the radical reading (or bounds κ); a null at the 10^{-3} level closes the last deviation channel and renders the mechanism empirically equivalent to standard quantum mechanics in every configuration tested.

8.6 Secondary predictions

For completeness: (i) *time-resolved Born* — click-time density $\propto |\text{envelope}(t)|^2$ (§6.1), already consistent with observation; (ii) *no speed floor* for the "speed of quantum influence" (§7.3) — a standing prediction distinguishing global from propagating resync; (iii) *polarization signature* — where array deviations exist, their sign rotates with polarization relative to the array axis (the vector vacuum kernels of §5.3: $c_\perp \approx -0.15$ vs $c_\parallel \approx +0.30$ at $\lambda/2$); (iv) *differential pre-coherence* — for the P3(a) channel, the active variable is the *difference* in pre-established coherence between ports, symmetric pre-coherence being protected by §8.2; the driven (local-oscillator) implementations of pre-coherence add quanta and exit the single-quantum sector, so their analysis requires the multi-quantum machinery of §6.3 and is left open; (v) *warm-detector bias* — **a candidate, not yet a ledger entry**: where site energies approach the bath's additive noise floor ($\hbar\omega \lesssim k_B T$; §4's companion-term analysis), the additive component drifts shares toward the leaders (Theorem 1), implying a reproducible odds *bias* toward bright sites over and above the excess dark counts — whereas standard quantum mechanics expects a marginal detector to grow noisier, not biased. Promotion to the ledger requires an analysis, not yet performed, separating this signature from dead-time, saturation, and afterpulsing systematics, and a quantitative treatment of the crossover region where signal stakes and bath floor compete. (vi) *energy-audit interferometry* — **a consistency test, not a discriminator, carrying a sponsor-originated interpretive clause**: send heralded single photons of known energy $\hbar\omega$ through a double slit or interferometer onto an *energy-resolving* detector plane (TES- or MKID-class calorimetry, few-percent resolution or better), and compare each click's deposited energy against the created photon's. Standard quantum mechanics and this mechanism predict the same result — the full quantum per click, at every fringe position, never a fractional deposit weighted by the local intensity — so a positive result discriminates *readings*, not theories: it directly excludes any local energy-accumulation picture in which sites keep their $|A_i|^2$ shares, complementing the first-click-latency exclusion of intensity-dependent delay, and it is the energy-domain twin of the spacelike anticorrelation already cited in §3. The mechanism adds one sharper clause — the sponsor's reading of Theorem 2: the distributed Stage-1/2 excitation is *reversible*, off-shell shares returned to the common field at κ_{ret}, so the energy "in the fringe" is never truly deposited; deposition completes only at registration, and losing sites should retain **no calorimetric residue** (no anomalous heating of dark fringes or of the losing arm) beyond ordinary optical losses. The canonical interference experiments could not ask this question — their click detectors are charge-triggered and energy-blind — and a literature search (2026-07) finds no published experiment framed as this audit; the closest existing evidence is incidental and large: energy-resolving MKID astronomical cameras (ARCONS, DARKNESS, MEC) record per-photon energy, position, and arrival time at focal planes shaped by diffraction, and TES calorimetry routinely confirms $\hbar\omega$ per absorbed heralded photon, while gamma-ray spectroscopy's photopeak/Compton structure is the same bookkeeping at MeV scale (full quantum at one site; any mismatch departing as real quanta). Promotion beyond consistency-test status requires the residue clause made quantitative — how much transient heating, if any, a calorimeter should register at a losing site during the selection window — which turns on the same κ_{ret} microphysics as §9.4(viii) and is logged with it. (vii) *Gap-ratio grading* — **a v0.8 candidate, derived from the corrected ladder of §6.1, not yet a ledger entry**: if the threshold-gated channel of Table 1 is real, the Born deviation at an asymmetric split is graded in $E_{\text{photon}}/E_{\text{gap}}$, through θ (the exposed fraction of the share axis) and through $\lfloor E_{\text{photon}}/E_{\text{gap}} \rfloor$ (how many real sub-gap excitations the pot can pay for). The clean test uses one detector family and one knob: a silicon SPAD pair behind a deliberately *asymmetric* splitter (symmetric ports are protected, §8.2), heralded single photons at two wavelengths straddling 553 nm, where $E_{\text{photon}} = 2E_{\text{gap}}$. Standard quantum mechanics predicts the same ratio, and $g^{(2)}(0) = 0$, at both. The exposed reading predicts a bright-favoured shift growing with $1-\theta$, and below 553 nm an excess-coincidence channel from the open two-pair possibility. Every downstream variable — avalanche, quench, readout — is held fixed by construction, which the SPAD-versus-SNSPD comparison the same ladder suggests cannot do. Promotion requires the discrete-exchange simulation of §9.4(xi) to give the shift a magnitude, and a literature check of asymmetric-split and $g^{(2)}(0)$ data taken with silicon SPADs below 553 nm. *(v0.9: both were done. The simulation puts reading A's deviation at of order* $+0.1$ *for asymmetric splits in silicon, not a percent-level shift; the search found no published asymmetric-split test below 553 nm, and found the coincidence branch bounded at* $g^{(2)}(0) = 0.0023$ *by heralded 505 nm data. Under* *the adopted reading B this test is predicted null; it discriminates the readings.)*

8.7 The cross-station injection test

The proposals of §§8.1–8.4 all target the single-detector premises. The entangled sector rests additionally on P5, and it admits its own instrument: *deliberately couple a common classical perturbation into both detectors of a working Bell experiment* — in the selection band, at the absorbers, bypassing the shielding by design — and read the coincidence statistics. Standard quantum mechanics predicts a null: correlated classical noise at the stations can move *rates* (dark counts, correlated efficiency drift — removable by fast setting-switching and time-tagged analysis) but cannot redirect *outcomes*, so the correlators are untouched. The present mechanism predicts a structured response, computed from the §7 joint game extended with cross-station noise [Framework repository: `DERIVATION_two_station_correlated_noise`, simulation suite `two_station_sims`], and the structure separates cleanly into two knobs.

Universal protections. Two results bound what any injection can do. First, each station's share drift is blind to cross-station covariance, so the *marginals stay exactly Born under any correlated injection whatsoever* — the effect, if any, lives only in coincidences, and the test is invisible in singles. Second, a *protection theorem*: for broadband (white) cross-station correlation of any strength $\rho < 1$, with faithful registry update ($\eta = 1$), the joint statistics remain exactly quantum. The correlation is genuinely generated — in simulation the far wing's share *leans* toward the near wing's developing outcome by $L = +0.29$ at $\rho = 0.8$ — and is then discarded, shot by shot, by the conditional reset of §7.1. White noise is additionally memoryless, so non-overlapping selection windows never see it at all. Broadband common-mode pickup between Bell stations is therefore harmless to the outcome statistics *even in-band*: a stronger statement than the engineering intuition that shielding suffices.

Knob 1 — broadband injection measures the registry fidelity. If the update is imperfect, the reset retains the fraction $(1-\eta)$ of the far wing's lean, and the leak is exactly computable: every correlator shifts by the same angle-independent amount, $\delta E = 2(1-\eta)L(\rho)$, hence

$$\delta S = 4(1-\eta)\, L(\rho),$$

confirmed by simulation at the percent level ($\delta S = +0.117 \pm 0.005$ observed against $+0.115$ predicted at $\eta = 0.9$, $\rho = 0.8$, with all four correlators shifted uniformly). Broadband injection at a *known, controlled* ρ therefore converts §7.2's inference ($\eta \gtrsim 0.99$ from existing CHSH values) into a laboratory dial: any measured δS is a direct reading of $(1-\eta)$.

Knob 2 — phase-locked injection and the above-Tsirelson signature. A coherent tone, phase-locked between the stations with global phase randomized shot to shot, is not white: its phase persists through the conditional reset and biases the far wing's *post-reset* game — the one contribution the registry cannot wipe. At leading order each correlator shifts by

$$\delta E(a,b) = 2\,\kappa_1\kappa_2\,\lambda^2 \cos 2(a-\psi)\cos 2(b-\psi),$$

with λ the injection's drift-to-diffusion ratio, ψ its polarization, and $\kappa_{1,2}$ the measured drift susceptibilities of the first- and second-committing wings. Three signatures follow, all verified in the simulated joint game: (i) $\delta S > 0$ — the tone *adds* classical common-bias correlation on top of the quantum correlation, pushing *S above the Tsirelson bound* $2\sqrt{2}$, which no quantum-mechanical mechanism can do; an $S > 2\sqrt{2}$ reading under injection would be an unambiguous non-QM signature. (ii) The angle-resolved pattern rotates with the injected polarization ψ — a fingerprint tying the deviation to the injection rather than to any systematic. (iii) The singles stay Born throughout (the shot-randomized global phase averages the marginal biases away), so the effect masks itself from every single-station monitor — deviation and no-signaling coexisting in exactly the way the two-stage construction prescribes.

The protocol carries its own calibration. The identical tone applied to *one* station with a *fixed* (not shot-randomized) phase is precisely the engineered-phase-reference limit of §4: it biases that station's singles by a computable amount, giving an in-situ measurement of the drift susceptibility product $\kappa\lambda$ at the absorbers themselves. The run order is therefore: single-station, fixed phase — calibrate $\kappa\lambda$ from the singles bias; then two-station, phase-locked, shot-randomized global phase — read the correlators. This closes the loophole that would otherwise drain a null result of meaning ("the injection never reached the absorbers"): with $\kappa\lambda$ independently measured, a two-station null bounds the *mechanism*, not the plumbing, and the tone-side prediction becomes falsifiable rather than merely confirmable.

Status and caveats. The tone-side prediction is presented at the same evidentiary level as §8.6(iv)–(v): the injected tone is modeled in the single-quantum sector as a share drift, whereas a real coherent field adds quanta and ultimately requires the multi-quantum machinery of §6.3; the ψ-invariance of δS at the standard angles is a leading-order statement (simulation shows comparable second-order response-curvature corrections at the tested $\lambda = 0.3$, with δS positive throughout); and near-simultaneous commits — within any finite commit duration the foliation cannot order — would escape conditioning at a rate that *grows* with cross-station correlation, a channel we flag as open. The broadband knob carries none of these caveats: the protection theorem and the leak law use only the §7 construction itself, and Knob 1 is proposed as a near-term, systematics-robust measurement of the one parameter (η) on which the entire entangled sector turns.

9. Discussion

9.1 In what sense is this a derivation of the Born rule?

Three answers, in decreasing strength of the question. In the strongest sense — probability extracted from a theory containing no probabilistic ingredient — it is not a derivation, and nothing ever will be: something must break ties, and every program in the literature derives the Born rule *from premises*, the real competition being over how few, how physical, and how independently motivated the premises are. In an intermediate sense it is a *conditional compatibility result*: the Born frequencies here follow as a theorem from premises that are physical statements about detectors, none of which mentions outcome probabilities, together with a frozen preparation measure over the detector's ready state. The squaring enters as oscillator energetics (P1); the randomness enters as ordinary noise (P2–P3); their conjunction is forced onto the unique fair point by proofs, not tuning.

But the sentence that stood here in earlier drafts — that "$|A|^2$ is nowhere assigned a probabilistic role by hand" — is withdrawn, and the boundary should be stated without softening. **This paper supplies, at most, the selector half of a larger theory.** It offers a candidate process that turns a distributed excitation into registration frequencies. It does not supply the measure those frequencies are compared against: the quantum outcome measure is adopted from the structural literature as a comparator, and the microstate preparation measure is an explicit premise with independently measured provenance. Neither is derived here, and agreement between the pushforward of the second and the value of the first is *compatibility across two different spaces* — not a demonstration that any equilibrium is dynamically preserved, and not a derivation of the Born measure. The other half of the theory — why the preparation measure is what it is, and whether a microstate flow preserves it — is untouched by anything in this paper. Two further debts belong in the same paragraph: P0 and P1 together carry the outcome weights into the argument as a premise (§1, §2), and the production of exactly *one* record — exclusivity, quench, and the routing of the losing sites' energy — is owed by §5's dynamics and is not discharged here. v0.9 sharpens the first of these: with reading B adopted (header), the Born frequencies enter through the linearity of the commitment hazard in the share — P1 together with Theorem 5 — and the exchange game's theorems state what the substrate must not do to the shares before commitment, rather than supplying the frequencies themselves; the substrate's own candidate for the actualization law is tested in the framework companion, not here. And in one specific respect the mechanism is *stronger* than the axiom it replaces: it specifies the physical conditions under which Born statistics fail (§8). An axiom cannot be wrong in instructive ways; a mechanism can, and twice in this work the mechanism's own theorems eliminated deviation channels that our first analysis had left open — a pattern we have displayed rather than hidden (Table 1).

9.2 Relation to other programs

Table 2 — Born-rule derivation programs compared.

Program	Selection mechanism	Noise physically identified	Fairness / measure status	Born statistics	Modifies quantum dynamics	Predicts deviations
Gleason (1957)	none (uniqueness theorem)	—	—	unique consistent measure	no	no
Deutsch–Wallace	none (rational credence)	—	—	derived as credence	no	no
Zurek envariance	none (symmetry argument)	—	—	derived from symmetry	no	no
Dürr–Goldstein–Zanghì	outcomes from beables; probabilities from typicality	—	equivariant measure, postulated typical	exact in equilibrium	no (adds beables)	no
Valentini	relaxation of the measure	—	equilibration dynamics	equilibrium value	no (adds beables)	**yes** (non-equilibrium)
Pearle / GRW / CSL	stochastic reduction	no	**postulated** (engineered noise)	exact	**yes** (new constants)	yes (heating, X-ray)
Hughston / ABBH	stochastic reduction	no	**postulated** (martingale by construction)	exact	yes	yes (weak)
S. Adler trace dynamics	emergent CSL	partially (substrate fluctuations)	postulated at effective level	exact	fundamental-level replacement	as CSL
Khrennikov PCSFT	threshold detection	yes (classical random field)	—	approximate	replaces QM with field theory	yes
Allahverdyan–Balian–Nieuwenhuizen	apparatus phase transition	yes (apparatus bath)	— (weights from QM)	inherited	no	no
This work	detector energy game	**yes** (vacuum/environment noise at the detector)	**derived** (Theorems 1–5)	**exact**	**no**	**yes** (specified family, §8)

Distinction from Everett / Many Worlds. Both Everettian quantum mechanics and the present framework retain unitary evolution and use detector–environment entanglement to explain the practical disappearance of interference. The distinction is ontological, not an alleged violation of energy conservation by Everett: Many Worlds treats the mutually decohered record branches of the universal wavefunction as equally actual, while the present program seeks one actual material registry or configuration throughout the same unitary cascade. Phase information is dispersed into system–detector–environment correlations and becomes locally inaccessible, but it is not destroyed or exported to additional worlds; information remains in the enlarged state, and the closed field–matter–bath–circuit energy ledger is counted once. Correspondingly, the framework should not say that Everett literally creates duplicate energy. Its positive claim is instead **no multiplication of actuality**: one physical record occurs, one ledger is realized, and the apparent termination of superposition reflects relational phase inaccessibility plus one-world selection rather than branching actuality.

This distinction creates a precise obligation. If strict unitarity is retained, the unselected amplitudes cannot simply vanish; a one-world beable/registry account must define their physical efficacy and stress–energy and explain why they cannot seed additional records, without booking their expectation values as separately possessed energy on top of the actual quantum transfer. If the losing amplitudes are instead dynamically driven to zero, the proposal is a continuous objective-reduction or winner-routing theory and is no longer strictly unitary at that level. The paper leaves this fork explicit rather than hiding it in the word "decoherence."

Three neighbors deserve individual comment. **Valentini's** quantum non-equilibrium is the nearest analogue of our deviation structure: there too the Born rule is an equilibrium condition that suitable physics could violate, with relaxation explaining why violations are unobserved. The difference is locus — relaxation of the configuration-space measure there, fairness of a detector-level game here — and consequently the predicted deviation signatures differ entirely (primordial/exotic matter there; engineered detector asymmetries here). **Khrennikov's** threshold-detection program is the nearest detector-side analogue: classical random fields plus threshold detectors yielding Born statistics. It obtains Born approximately and calibration-dependently, without the winner-take-all competition; the martingale structure is what buys exactness here. **Allahverdyan, Balian and Nieuwenhuizen's** Curie–Weiss analysis is the nearest apparatus-side analogue: registration as a metastable apparatus tipping into one basin, treated with full statistical-mechanical care within unitary quantum theory. It takes the outcome weights from standard quantum theory; the present work supplies the selection stochastics beneath such weights. We regard the three as complementary flanks of the same position — that measurement is detector physics — approached from the state, the field, and the apparatus respectively; this paper approaches it from the noise.

9.3 Relation to a broader program

The premises of §2 are not free-floating. They are realized in a specific substrate framework — the Dirac–Kuramoto program — in active development by the present authors, which supplied the wave ontology behind P1, the coupling postulate P2, the locking threshold of P4(i), and the shared registry and foliation of P5–P6, and which motivated this investigation. The framework's development is publicly documented in a version-controlled repository [Framework repository], whose complete, timestamped revision history constitutes the provenance record of the program, including the evolution of the present mechanism through its intermediate (and twice-retracted) forms. A companion paper develops the measurement-structure consequences — the classical–quantum boundary as the locking threshold quantified in Theorem 2. The present paper is nonetheless deliberately self-contained: its results are conditional on the stated premises and stand or fall with them, not with the framework; any ontology supplying P1–P6 inherits every theorem.

Relational-phase interpretation. Absolute or common phase is not directly observable; experiments reveal phase only by comparison with another path, oscillator, apparatus, or reference field. The framework accordingly interprets measurement as a joint evolution in which the detected system, detector, environment, and ultimately the observer's physical records become correlated with a common phase history. Continued rotation shared by the system and its effective reference can then be locally invisible: the relative phase available to that observer is stationary or dispersed into inaccessible environmental correlations. On this reading, superposition need not literally stop evolving when interference disappears. What looks like collapse can instead include phase entrainment plus decoherence — continued global evolution whose branch-relative phase is no longer operationally accessible from within the correlated record.

In this account, "dispersion" means that phase and information become encoded nonlocally in correlations across the detector and environment, not that energy or information is destroyed. Unitary evolution preserves the global information, while ordinary energy currents distribute the conserved real energy among electromagnetic, electronic, phonon, circuit, and radiative degrees of freedom. The observer generally lacks control of enough correlated degrees of freedom to reverse the cascade or measure the original relative phase.

This interpretation must be kept separate from outcome selection. Standard unitary decoherence already explains why interference between macroscopic records becomes locally inaccessible while the global superposition persists; common phase rotation alone does not explain why one record is actual rather than an improper mixture of alternatives. The framework still owes either a one-world beable/registry law with a derived Born distribution and global exclusivity, or an explicit continuous winner-routing law with conserved field–matter–bath currents. The relational-phase picture is therefore a proposed account of **apparent termination and phase inaccessibility**, not by itself a derivation of **single actuality**.

"Kept separate" here means separated as causal stages, not disconnected explanations. The proposed mechanism joins them in order: detector–environment coupling converts initially coherent alternatives into bath-defined channels; relational phase becomes inaccessible from within the correlated record; the fair selection dynamics supplies the additional one-world transition; and commitment plus amplification stabilizes that selection as a macroscopic observation. This division lets decoherence do the job unitary dynamics demonstrably supports while assigning the unresolved actuality problem to a specific dynamical layer rather than to an observer's update.

9.4 Open problems

(i) *The fork of §8.5*: whether a deeper fairness principle closes the mismatched-port channel — the mechanism's most consequential open question, and experimentally addressable (§8.4). (ii) *Deriving* $\eta = 1$: P5's faithful renormalization is measured to hold at the percent level; the frame-rotation picture of §7.3 suggests why it should be exact, but a dynamical derivation is lacking. (iii) *Multipartite extension*: the GHZ configuration, where the two-stage construction must iterate. (iv) *Deriving P1 microscopically* within a concrete relativistic oscillator model, replacing generic energetics; and promoting the interference-cross-term origin of P2's noise scaling (§4) from scaling argument to a controlled open-system calculation — with conservation enforced and the companion terms (heating, additive floor, damping) tracked — which would make the fair noise a *result* of surface wave mechanics rather than a premise about it, with the fairness window $\hbar\omega \gg k_B T$ of §4 expected to emerge as the calculation's domain of validity. (v) *The dressed-mode comparison* for radiative-dominated arrays, required before the §8.3 channel's magnitude can be trusted. (vi) *Formalizations*: general-N induction to continuous-mode Glauber theory; arrays mixing detector classes; second-order corrections (slaved-site correlations; the small equalizing residual in strongly correlated matched ports). (vii) *The injection test's open edges* (§8.7): the multi-quantum treatment of the phase-locked tone; the finite-commit-duration ordering window, whose escape rate grows with cross-station correlation; and the small-λ regime where the leading-order tone law becomes quantitatively clean. (viii) *The return-rate ansatz* (escalated from the companion measurement-structure paper's review panel, finding N2): derive $\kappa_{\text{ret}} = \Delta E/\hbar$ — and the legitimacy of modeling reversible give-back as a contraction eigenvalue in the site's reduced description — from an explicit $H = H_{\text{field}} + H_{\text{absorber}} + H_{\text{bath}} + H_{\text{int}}$, identifying the actual reduced-dynamics poles; and settle, in the same calculation, whether vacuum coupling restores an effective free phase below the classical threshold.

(ii) *Relational-phase completion*: formulate the "observer shares the phase rotation" claim using explicit system, apparatus, environment, and reference-frame degrees of freedom; identify the gauge-invariant relative-phase observables; and show which coherences become inaccessible under entrainment and decoherence. The same model must state what, beyond phase inaccessibility, selects a unique registry value and prevents a second separated detector from committing to the same single quantum. Without that additional result, the phase account explains apparent collapse but not single actuality.

(iii) *Sector-blindness of the registry update*: establish whether the substrate's update mechanism treats a delocalized single-quantum amplitude and a two-quantum pair registry alike. If it does, the conservation argument that forces $\eta = 1$ in the single-quantum sector (P5, §7.2) carries to the entangled sector and removes η as a free parameter there; if it does not, the two sectors need independent justification and the fidelity law of §7.2 stands on its own.

(iv) *Share ontology and the exposed channel* (v0.8): whether a partial share above θ can form a real excitation at all. §6.1's protection argument presupposes that shares are energies which would decay if they could; if instead they are off-shell amplitude bookkeeping that becomes energy only at $s = 1$ — Theorem 2's picture for class (i), extended to class (ii) by P4(a) — then no exposed channel exists, the top rung of the ladder is not a timescale but the premise P4(a), and the gap-ratio grading of §8.6(vii) dissolves. The paper cannot hold both readings; the open-system calculation of (iv) and (viii) is where they separate. Owed alongside: a discrete-exchange simulation with the commit channel gated at θ — rate $\propto e_i$ and, separately, Arrhenius — swept over θ and over the marginal range $t_{\mathrm{exp}}/\tau_{\mathrm{commit}} = 10^{-2}$–$10^1$, replacing the one-step diffusive estimate of §6.1 with a curve. *(v0.9: done — `g3_drain_tests/threshold_gated_commit.py`; exposure is 40–90% of the game, a gate biases even the linear law, and reading B is adopted as the working choice (header). The fork is kept because A remains testable at asymmetric splits below 553 nm. What is now owed in its place is the actualization law: a substrate dynamics whose first-firing hazard is linear in the share without that linearity being inserted — the Adler-locking race of the framework companion. Its first diagnostic-budget run (2026-09-02; header) gives an exponent of* 1.56 [1.44, 1.69] *at the frozen criterion, criterion-dependent: the race as specified does not yet supply that law.)*

The one-sentence summary of the paper is this: with the noise physically identified and the fairness argued from detector premises rather than engineered, the Born comparator is reproduced by a fair fight over deposited energy, in one world, with the odds set by mechanics and the fights we could rig identified in advance — a candidate selector, conditional on a declared preparation measure and on premises P0–P1, not a derivation of the measure it reproduces and not yet a demonstration that the fight ends with exactly one winner.

Appendix A — Share drift under general noise: proofs

A.1 General noise law, independent sites. Let $de_i = \sigma\, a(e_i)\, dW_i$ with independent Wiener increments. Itô's lemma on $s_i = e_i/E$, using $\partial s_i/\partial e_j = \delta_{ij}/E - e_i/E^2$ and $\partial^2 s_i/\partial e_j^2 = -2\delta_{ij}/E^2 + 2e_i/E^3$, gives

$$\mathbb{E}[ds_i] = \frac{\sigma^2}{2}\sum_j a(e_j)^2\,\frac{\partial^2 s_i}{\partial e_j^2}\,dt = \frac{\sigma^2}{E^2}\Big[\,s_i\sum_j a(e_j)^2 \;-\; a(e_i)^2\Big]\,dt.$$

This vanishes for *all* configurations iff $a(e)^2 \propto e$ — the amplitude-linear law — proving Theorem 1 and its uniqueness. The two failure laws quoted in §5.1 are immediate: $a = a_0$ (additive) gives drift $(\sigma^2 a_0^2/E^2)(Ns_i - 1)$, positive for above-average sites; $a = e$ (multiplicative) gives $(\sigma^2 e_i/E^2)(\frac{1}{E}\sum_j e_j^2 - e_i)$, negative for the leader.

A.2 Correlated noise. With $a(e) = \sqrt{e}$ and $\langle dW_i\, dW_j\rangle = C_{ij}\, dt$, the mixed second-derivative terms $\partial^2 s_i/\partial e_j \partial e_k = -(\delta_{ij} + \delta_{ik})/E^2 + 2e_i/E^3$ contribute and the same computation yields the formula of Theorem 3:

$$\mathbb{E}[\dot{s}_i] = \frac{\sigma^2}{E^2}\Big[s_i(\sqrt{e}\cdot C\cdot\sqrt{e}) - \sqrt{e_i}\,(C\sqrt{e})_i\Big].$$

Checks: $C = \mathbb{1}$ recovers A.1's zero; $C_{jk} \equiv 1$ (common noise) gives, for $N = 2$, $\mathbb{E}[\dot{s}_1] = (\sigma^2/E^3)\sqrt{e_1 e_2}\,(e_1 - e_2)$ — the rich-get-richer drift quoted in §5.3; linearity in C underlies the mixed-kernel scaling (deviation $\propto f$) exploited in §5.3 and §8.

Appendix B — The slaved phase: linear analysis and layer width

B.1 The sub-threshold mode. In a frame rotating at the drive frequency, with the detuning convention $\dot{a} = -(\kappa_{\mathrm{ret}} + i\Delta)\, a + f e^{i\varphi_d}$ and $\kappa_{\mathrm{ret}} = \Delta E/\hbar$ the off-shell return rate (§5.2 dictionary) (P1, P4(i)), the system is linear and inhomogeneous. Its unique fixed point is $a^* = f e^{i\varphi_d}/(\kappa_{\mathrm{ret}} + i\Delta)$, i.e. $r^* = f/\sqrt{\kappa_{\mathrm{ret}}^2 + \Delta^2}$, $\varphi^* = \varphi_d - \arctan(\Delta/\kappa_{\mathrm{ret}})$; the homogeneous eigenvalues are $\lambda = -\kappa_{\mathrm{ret}} \pm i\Delta$, so the flow is a global contraction. No trajectory winds: phase deviations decay at rate κ_{ret}, and the stationary phase offset is a function of $(\kappa_{\mathrm{ret}}, \Delta)$ *only* — identical for identical sites, independent of stored energy. By contrast, the R. Adler injection-locking equation $\dot{\varphi} = \Delta - K\sin\varphi$ [R. Adler 1946] presupposes an *autonomous* oscillator: amplitude frozen at a limit-cycle radius maintained by the oscillator's own gain, phase a neutrally stable variable free to run. Its running solutions (for $|\Delta| > K$) carry the nonzero average $\langle \sin\varphi\rangle = [\Delta - \mathrm{sgn}(\Delta)\sqrt{\Delta^2 - K^2}]/K$ — the entrainment pull. Below threshold there is no gain, no limit cycle, no running solutions, and hence no pull: the rich-get-richer channel is structurally absent, which is Theorem 2.

B.2 Layer width, two ways. (i) *Marginality:* the phase becomes exploitable when its restoring rate falls below the coupling, $\kappa_{\mathrm{ret}} \lesssim K$, i.e. $\Delta E \lesssim \hbar K$, giving fractional width $w = K/\omega$. (ii) *Width–commitment* *identity:* the same Γ that broadens the final state is, by fluctuation–dissipation, the rate of irreversible decay into registered products; with $K \sim \Gamma$ this reproduces $w = \Gamma/\omega$. The two estimates agree because they are the same physics seen from the dynamical and the dissipative side.

B.3 Feedback hierarchy. From $\dot{e} = 2r\dot{r} = -2\kappa_{\rm ret}, e + 2f\sqrt{e}\cos(\varphi_d - \varphi)$: uptake scales as $e^{1/2}$, decay as e^1. For two sites exchanging via any channel with uptake $u(e) = c\,e^p$, the share drift obeys $\dot{s}_1 \propto u(e_1)e_2 - u(e_2)e_1 = e_1e_2[u(e_1)/e_1 - u(e_2)/e_2]$, whose sign for $e_1 > e_2$ is that of $p-1$: concave uptake ($p < 1$) equalizes, linear ($p = 1$) is share-neutral, superlinear ($p > 1$) amplifies. The slaved sector contains only $p = \frac{1}{2}$ (equalizing, toward the drive pattern — which by P1 is the Born pattern) and $p = 1$ (neutral); the amplifying sector requires autonomy and is confined to the layer.

Appendix C — Kinematic linearity of registration (Theorem 5)

Registration is a transition from the delocalized n-quantum excitation to a final state containing a registered excitation of energy $k\hbar\omega$ at site i (plus $n-k$ quanta remaining). At leading order in the registration coupling, the T-matrix element consists of k absorption vertices at the site; each vertex is amplitude-linear (P2), so $M_i \propto \psi^{(k)}(i,\dots,i)$ — one factor of the k-quantum joint amplitude — and the golden rule gives rate$_i \propto |\psi^{(k)}(i,\dots,i)|^2 = G^{(k)}(i)$: linear in the coincidence weight, never a higher power. The alternatives are eliminated as follows. *Fewer than k vertices:* the energy deficit has no compensating final state (P4(i): no sub-gap levels), and thermal compensation is Boltzmann-suppressed ($e^{-\Delta E/k_BT} \sim e^{-10}$ already at $\Delta E = 0.25$ eV, room temperature). *More absorption vertices:* $(k{+}1)\hbar\omega$ final states are absent in-sector, or constitute a different channel, itself linear in its own $G^{(k+1)}$. *Absorb–emit insertions:* return quanta to the pool — they are the exchange dynamics of §4, not registration; their leading contribution to the commit channel is a three-vertex path whose amplitude is suppressed by two powers of (coupling/energy denominator), a parametric estimate $(\Gamma/\omega)^2 = w^2 \lesssim 10^{-4}$ for broadband silicon and $\lesssim 10^{-14}$ for atomic lines. *Saturation:* occupation factors $(1-n_f)$ equal unity for a ground-state detector. Combining with the affine ensemble structure (§6.3): every commit channel is linear in a $G^{(k)}$, every $G^{(k)}$ is linear in ρ, per-configuration statistics are Born (§5), and mixtures average affinely — all outcome statistics are affine in ρ, in every sector.

Appendix D — Simulation methods and code availability

All simulations use a common stochastic engine: Euler–Maruyama integration of $de_i = \sigma\, a(e_i)\,\Delta W_i$ with $\Delta t = 0.02$, $\sigma = 0.3$ (units $E_0 = 1$), and clipping at zero (for the $\sqrt{e}$ law, zero is naturally sticky — a CIR-type boundary — implementing drop-out); the discrete gambler variants use stakes-scaled exchange $\delta = \pm\,\text{step}\cdot\min(e_i, e_j)$ with fair sign. Registration is implemented either as a share threshold (absorption at $s \geq 0.7$–0.995 as noted per experiment) or as a rate process (geometrically distributed commit times; conditional pick $\propto s^\alpha$). Correlated noise uses covariance factorization by eigendecomposition with eigenvalue clipping. Ensemble sizes are 1.5×10^3–6×10^4 trials per data point, with Monte-Carlo floors quoted alongside results in the text. Deliberately *failing* configurations (clipped fixed-step exchange; additive and multiplicative noise laws; extreme-value selection; biased-layer profiles; common-mode noise; nonlinear commit rates) are retained in the code base as diagnostics: each failure realizes a specific theorem's converse, and their outputs are reported in §§4–8 with the same prominence as the passing runs. The complete simulation suite (thirteen scripts, all random seeds pinned, outputs matching every number quoted in this paper) is archived with an index mapping each script to the results and figures it supports [repository ref]. The two-station injection-test simulation of §8.7 (correlated Wright–Fisher joint games with the §7.1 conditional reset; same-step double commits ordered explicitly, without which correlated paths silently skip conditioning and mimic η-degradation) is archived separately as `two_station_sims/` with its full-statistics output.

Appendix E — Closed forms for hypothetical deviations

E.1 Signaling amplitude of a commit nonlinearity (§7.4). For fast commitment with pick probability $\propto s^{1+\varepsilon}$ at wing A, state $\sqrt{p}\,|00\rangle + \sqrt{1-p}\,|11\rangle$, Alice at angle α: her local shares are $s_+ = p\cos^2\alpha + (1-p)\sin^2\alpha$,

$s_- = 1 - s_+$, and Bob's conditionals are $P(B{=}0 \mid A{=}\pm) = p\cos^2\alpha/s_+,\ \ p\sin^2\alpha/s_-$. Composing,

$$M(B{=}0;\alpha) = p\,\frac{\cos^2\alpha\ s_+^\varepsilon + \sin^2\alpha\ s_-^\varepsilon}{s_+^{1+\varepsilon} + s_-^{1+\varepsilon}},$$

which is α-independent ($= p$) iff $\varepsilon = 0$. Expanding at $\alpha = 0$ against the fixed point $M(45^\circ) = p$:

$$\Delta M = \varepsilon\, p(1-p)\ln\frac{p}{1-p} + O(\varepsilon^2), \qquad K(p) \equiv p(1-p)\ln\frac{p}{1-p},$$

with $K(0.8) = 0.222$ at leading order (effective $K \approx 0.18$–0.21 over the simulated ε range), halved in the full process by $P_{\text{first}} = \frac{1}{2}$. Both exact zeros — $\varepsilon = 0$ and $p = \frac{1}{2}$ — are manifest in the formula.

E.2 Port-level drift and the κ model (§8.2–8.3). Summing A.2 over the sites of port A (block-structured C; cross-port correlations zero) gives

$$\dot{S}_A = \frac{\sigma^2}{E^2}(q_B S_A - q_A S_B), \qquad q_P = \sum_{i,j\in P} C_{ij}\sqrt{e_i e_j},$$

vanishing iff $q_A/S_A = q_B/S_B$ (matched collective structure). For the uniform-port model — m equal-energy sites per port, internal correlation c_P — one finds $q_P = S_P E\,[1 + (m-1)c_P]$ and hence

$$\dot{S}_A = \frac{\sigma^2}{E}\, S_A S_B\, (m-1)\,(c_B - c_A):$$

drift proportional to the collective-structure mismatch, suppressing the internally-correlated port, and vanishing for matched ports at any c. A constant drift-to-diffusion ratio integrates to the biased-ruin form $P'(S) \approx S + \kappa S(1-S)$ quoted in §8.3, with $\kappa \propto (m-1)(c_B - c_A)$; the simulations give effective $\kappa \approx -\Delta c_{\text{eff}}$ at $m = 4$ over the tested range.

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